Conducting Qualitative Interviews with AI

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Abstract

Qualitative interviews offer rich insights into human behavior, but their lack of scalability has limited their use in economics. We develop the first scalable method for conducting qualitative interviews by delegating the task of interviewing human subjects to an AI interviewer. We then conduct a comprehensive quality evaluation of AI-led interviews. The AI interviewer follows methodological guidelines and achieves high interviewee satisfaction. AI-led interviews also allow hypothesis generation, outperform standard open-ended survey questions with richer discoveries, and predict economic behavior half a year later. Our findings demonstrate that AI-conducted interviews can generate rich, high-quality interview data at a low cost.

Keywords: Artificial Intelligence, Interviews, Large Language Models, Qualitative Methods, Mental Models, Surveys.

JEL codes: C83, C90, D14, D91, Z13

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1 Introduction

Understanding human behavior requires qualitative data on how people perceive and reason about the world. Economists are increasingly turning to qualitative data to study many decision-relevant factors that are often unobserved in traditional datasets, such as narratives (Andre et al., 2023), mental models (Andre et al., 2024), reasoning (Stantcheva, 2021), attention (Link et al., 2024), or memory (Graeber et al., 2024b). Qualitative interviews are a leading method for collecting qualitative data in other social sciences, allowing researchers to ask follow-up questions to better understand the participant's perspective (Knott et al., 2022). Yet, interviews have seen limited adoption in economics, as their labor-intensive nature makes them impractical for economic research designs that require scale. Artificial intelligence (AI) promises an opportunity to overcome these constraints by delegating the task of interviewing human subjects to an AI interviewer. However, a key open question is whether AI-led interviews can generate high-quality qualitative data that provide significant benefits over traditional survey methods.

In this paper, we develop the first scalable AI interviewer and conduct a comprehensive examination of the quality of AI-led interviews with human subjects. Evaluating interview quality is challenging as there is no broad consensus among qualitative researchers on how to evaluate the quality of qualitative data. We address this challenge with four complementary strategies: (i) evaluating the interview process, including participants' evaluations and a human evaluation of interview questions; (ii) assessing whether AI-led interviews allow *hypothesis generation*; (iii) benchmarking AI-led interviews against existing *survey methods* for gathering qualitative data, and (iv) showing that interview responses are internally and predictively valid for studying human behavior. We first introduce our AI interviewer method before presenting the findings from our evaluations.

Our scalable AI interviewer application prompts a frontier large language model (LLM) to dynamically generate questions for an interview with a human subject in real time. The key input for the LLM is a short prompt with detailed guidelines on interviewing derived from the literature. For example, the prompts explain that it is central to ask neutral, open-ended questions instead of narrowly framed questions that prime respondents to answer in a specific way. The core prompts do not include application-specific instructions, making them portable across contexts. To conduct interviews on a new topic, researchers have to write down a short *topic guide* that specifies the length and scope of the topics that the AI interviewer should ask about, giving the researcher full control over the interview structure. Our application is modular and includes many optional features, such as (i) content moderation of interviews to detect problematic messages (Jiménez-Durán, 2021), (ii) generating on-the-fly analytic memos to help the LLM focus its attention on important details in very long interviews, and (iii) having separate LLMs for different types of interview questions to benefit from task separation (Khot et al., 2023). Participants interact with the AI interviewer through a chat interface that can be

seamlessly embedded into standard survey software.[1] The chat interface is programmatically protected against participants inserting AI-generated text. To answer an interview question, participants can either type their response or record a voice message that we automatically transcribe. Within a second, the next interview question is then shown to participants. Our application is open-source with production-ready code that can be deployed at zero cost within the free tier of standard cloud infrastructure.[2] With LLM usage costs below $0.10 per interview hour, our method enables researchers to run unlimited interviews in parallel at effectively zero marginal cost compared to a standard survey.

To empirically evaluate our method, we conduct 766 AI-led online interviews with US Americans that lasted for close to 400 hours in total and generated 1,026,000 words of interview transcripts. The data collection required about ten days in the field, and is thus substantially faster than traditional interviews. We selected two complementary applications that account for a large share of economic research, i.e., understanding (i) economic behavior and (ii) economic policy views. Specifically, for demonstrating the value of AI-led interviews for studying economic behavior, we interviewed non-investors about their reasons for not participating in equity markets in August 2023. For our second application, we interviewed US Americans about their views on US import tariffs in May 2025. These applications provide ideal settings for evaluating AI-led interviews because of the significant heterogeneity in expected responses and the likely discrepancy in reasoning between laypeople and economists on these topics, which renders rich open-ended data especially valuable.

Interview process quality Before turning to the thematic content of our interviews, we begin our evaluation of interview quality by examining the interview process itself. As AI-led interviews are conducted without direct researcher involvement, we cannot rely on anecdotal observation to assess what the AI interviewer is doing or how participants are responding. Furthermore, it is a common problem that LLMs often fail to follow explicit instructions (Zhou et al., 2023), making a systematic evaluation important. To open this black box, we systematically hand-code over 18,000 interview questions and responses for ten markers of interview quality. This task requires a human annotator to avoid circularity, as even frontier LLMs are known to have low accuracy in identifying their own mistakes (Kamoi et al., 2024). First, we find that the AI interviewer robustly adheres to key methodological principles for interviewing. For example, over 95% of questions are rated as open-ended rather than narrowly framed, perceived as neutral and non-leading rather than suggestive of any possible response, and deemed highly relevant to the interview topic. Concerns about the AI interviewer biasing responses are thus not supported by the data. Of course, interviewing requires more than simply asking open-ended questions. A

[1] Experimental evidence suggests there are no differences in the quality and quantity of information communicated in online interviews compared to face-to-face interviews (Namey et al., 2020). An online interview might have several benefits on its own, including a greater sense of privacy and control (Gibson, 2022). Whether face-to-face conversations are needed depends on the researcher's objectives.

[2] The code, instructions and setup scripts are available here: `https://github.com/fchop/interviews`

skilled interviewer responds to conversational cues, clarifies ambiguous points, and follows up on notable themes. To evaluate these aspects, participants' responses are rated on their ambiguity and need for further clarification. We find that the probability of asking a clarification question strongly increases after ambiguous statements. This suggests a robust contextual understanding of what types of questions are needed. Reassuringly, we also find essentially zero instances of hallucination.

Second, we use participants' own evaluation of the interview as a direct lens into the interview quality. Participants consistently (i) rate the overall interview as very positive, (ii) perceive the conversation as feeling very natural, (iii) perceive the questions as open-ended and unbiased, (iv) indicate that the interview allows them to accurately express their own views and opinions about the topic, and (v) report feeling comfortable during the interview. As a result, a majority of participants would prefer an AI interviewer over a human interviewer and over 96% would like to participate in another AI-led interview. These evaluations are consistent with other evidence. First, when hand-coding transcripts, we detect almost no signs of discomfort or annoyance with the interviewer. Second, participants exert consistently high effort when answering questions, without any decrease in the amount of text produced or the time taken to answer questions over the course of the interview. Third, we replicate a weak preference for AI interviewers in a separate survey experiment in which we randomize whether participants are offered the opportunity to participate in an AI-led or a human-led interview, thereby addressing potential concerns about selection.

Hypothesis generation We next turn to the *content* of interviews and the hallmark of qualitative research: the generation of hypotheses grounded in participants' own accounts. Although economists traditionally focus on hypothesis testing, data-driven hypothesis generation is increasingly seen as scientifically valuable (Ludwig and Mullainathan, 2024). We therefore show that AI-led interviews can generate novel insights into human behavior, thus demonstrating their usefulness for classical qualitative research purposes. For this exercise, we conduct an inductive thematic analysis of our interview transcripts with the goal of discovering notable themes *from the data itself* rather than forcing categories onto the data from a priori theory (Glaser and Strauss, 2017). We borrow techniques from computational sociology to show that the analytical cost for the researcher can be reduced, while maintaining full human control over the interpretation of the raw data (Nelson, 2020). The output of this analysis is a coding manual with a set of codes representing notable themes in the data. We then hand-code transcripts based on the coding manuals and empirically validate that this task can be delegated to an LLM at low cost by benchmarking AI coding against human annotators.

In both applications, we recover a wide range of systematic themes, including both novel aspects and those consistent with previous research. Perhaps most interestingly, a recurring finding across our applications is that AI-led interviews uncover novel *mental models* and economic mechanisms shaping people's views about the world. In the tariff interviews, for

example, we find that disagreement between supporters and opponents is not necessarily about whether tariffs have costs—nearly all participants acknowledge that tariffs raise prices in the short term—but about how they reason through the economic mechanisms shaping longer-term outcomes. Opponents, for example, point to structural barriers and the competitive advantages of other countries to explain why tariff-induced job creation is unlikely. Supporters, by contrast, argue that consumer substitution effects and innovation incentives will mitigate adverse economic damages and render the US economy more competitive in the long run. Our interviews thus help identify the causal reasoning underlying people's policy views.

Similarly, in the stock market interviews, we find that many participants hold a mental model that leads them to overestimate the actual cost of participating in equity markets. Participants believe that successful investing requires *actively* selecting individual stocks, constant portfolio monitoring, and quick decision-making in response to ever-changing market prices. As a result, people mistakenly conclude that owning stocks is necessarily a time-consuming and stressful activity that is not justified by the expected returns. Our interviews thus provide a strong argument for the relevance of misperceived participation costs in driving non-participation.

We then demonstrate the external validity of this discovery out-of-sample. Specifically, we conduct a separate survey ($n = 1,000$) with investors and non-investors that includes tailored survey questions to directly measure the presence of this mental model. Reassuringly, we find that the mental model is 22 percentage points more common among non-investors. This exercise thus provides a proof of principle that AI-led interviews can generate meaningful insights into economic problems that can be validated out-of-sample with standard survey techniques.

Benchmarking against survey methods A central goal of this paper is to examine whether AI-led interviews improve on the status quo in economics of using open-ended survey questions to collect qualitative data. This comparison between surveys and AI-led interviews is particularly relevant as the two modes of collection have an almost identical cost structure. Our goal is thus not to replace human-led qualitative interviews, but rather to examine whether AI-led interviews allow survey researchers to collect better qualitative data at scale.[3] To answer this question, we conduct separate benchmark surveys. First, we compare AI-led interviews against the common approach of asking a single open-ended survey question. Second, we design a series of predetermined open-ended questions that mirror the structure and content of the interviews. These mirror questions thus account for the effect of simply asking more open-ended questions, thus isolating the value of asking tailored follow-up questions. We then use the same coding manual to analyze both the AI-led interview transcripts and the "transcripts" obtained by combining the responses to the mirror survey questions.

We obtain three central results. First, AI-led interviews significantly increase text production on a question-by-question basis. Participants in an AI-led interview write up to 92% more

[3] Surveys are the preferred method for measuring ex-ante known, well-defined constructs. Our focus is therefore on whether interviews outperform surveys at the task of gathering open-ended, qualitative data.

text on average, even when comparing answers to conceptually identical questions. Moreover, participants using our voice input feature further increase their text output by another 112%. This suggests that AI-led interviews overcome the key challenge of short responses that often make it difficult to analyze responses to open-ended survey questions. Second, AI-led interviews discover a broader set of codes than open-ended survey questions. Across applications and benchmarks, we find that between 46% and 84% of the recurring themes observable with an AI-led interview remain undetected when using text responses to standard open-ended survey questions. At the participant level, a single open-ended survey question yields about two assignable codes per participant. By contrast, an AI-led interview increases the number of unique codes per participant to more than ten, with the average frequency of codes doubling to between 31% and 35% across applications. Third, we examine what exactly is being missed by survey questions. We find that even when asking multiple open-ended questions, survey responses mainly capture a narrow range of concerns that are top-of-mind, struggling to reveal *why* and *how* people reason. By contrast, across our applications, we find that AI-led interviews reveal conceptually novel *classes* of codes, such as mental models, causal narratives, and economic mechanisms, and can uncover contextually grounded themes such as social influences and personal experiences that rarely surface in responses to survey questions. Taken together, these findings suggest that AI-led interviews provide a "double dividend" by increasing the breadth of themes observable to researchers both at the participant and at the aggregate level.

Internal and predictive validity Finally, we address the old critique of survey methods in the context of AI-led interviews: Do people mean what they say? In other words, how much can we trust interview responses to be genuine? We address this concern in three ways. First, we show that what people say during interviews strongly correlates with their responses to validated survey instruments designed to elicit the same construct (e.g., Manski, 2004; Falk et al., 2018). For example, participants score half a standard deviation higher on survey measures of risk-aversion if risk-related concerns surface during interviews. Second, we show that there is a remarkably high degree of internal consistency between what participants say during interviews and how they respond to thematically related multiple-choice and Likert scale questions elicited later in the survey. For example, multiple choice options in the survey are about 50 percentage points more likely to be selected if a matching theme surfaced during the interview. Third, we show that what people say during interviews predicts incentivized choices more than half a year later. Specifically, we conduct a long-term follow-up survey with our stock market interview sample after eight months. The follow-up includes incentivized outcomes, such as willingness-to-pay measures or portfolio choices, that we show to be strongly correlated with participants' interview responses in the expected way. For example, participants who state they are unwilling to make nominal losses in the interviews are 180% more likely to demand 100% loss protection to put $1,000 at risk in the stock market. We are not aware of a comparable study that demonstrates the predictive validity of interview responses for future behavior. This evidence suggests that AI-led

interviews capture meaningful factors that reflect how people actually think and behave.

Contribution This is the first paper to propose, develop, and systematically evaluate a scalable method of interviewing human subjects with an AI. Our main contribution is to demonstrate that AI-led interviews allow researchers to collect richer qualitative data at scale. Our open-source tool makes it straightforward for other researchers to adopt our methodology, making it feasible to embed interviews into large-scale surveys at essentially zero marginal costs.

Our findings contribute most directly to the literature on survey methods in economics (Haaland et al., 2023; de Quidt et al., 2018; Stantcheva, 2023; Fuster and Zafar, 2023; Manski, 2004; Coibion and Gorodnichenko, 2026), which has long recognized the limitations of closed-ended survey questions (Lazarsfeld, 1944). A rapidly growing body of work has started to use open-ended survey questions to collect qualitative data from human subjects (Pinger et al., 2025; Andre et al., 2023, 2024; Stantcheva, 2021; Bursztyn et al., 2023b; Link et al., 2024). Another strand of the literature has started to explore the potential of audio data for survey research (Graeber et al., 2024c,a; Galasso et al., 2024). Yet, a key challenge for the literature is that survey responses tend to be short and tend to capture what is "top of mind" (Haaland et al., forthcoming). Our main contribution is to show that AI-led interviews allow survey researchers to collect richer qualitative data at scale without an increase in costs.

Our methodological innovation has sparked a rapidly growing literature developing open-source implementations of AI-led interviews and studying the properties of AI-led interviews.[4] Wuttke et al. (2024) conduct a small-sample experimental comparison of AI-led to human-led interviews and find similar response quality. Zarifhonarvar (2024) provides non-technical instructions on how to integrate AI interviewers into surveys. Geiecke and Jaravel (2024) replicate our finding that participants prefer AI-led to human-led interviews in different interview contexts using a single-agent framework. They also find that PhD students in sociology rate the AI interviewer to be comparable to a hypothetical average human interviewer. Finally, leveraging a multi-agent design directly inspired by our approach, Guven et al. (2025) randomize interviewees into interviews conducted by an AI or a human interviewer and find no significant differences in response quality between the two interview methods.

Finally, we contribute to classical and recent work in economics using interviews and qualitative data for hypothesis generation (Bewley, 1995; Shiller, 1997; Blinder et al., 1998; Bewley, 1999; Morduch and Schneider, 2017; Sivaramakrishnan et al., 2017; Duraj et al., 2024; Mutz, 2021; Bustos et al., 2022; Bergman et al., 2024; Baader et al., 2025, see Starr (2014) for a review), which is increasingly recognized as a key scientific activity by economists. In related work, Ludwig and Mullainathan (2024) propose a formalized method for generating hypotheses

[4]Related work uses conversations with chatbots as treatments (Ash et al., 2025; Costello et al., 2024) or as substitutes for hiring interviews (Jabarian and Henkel, 2025). More broadly, we relate to work examining the impact of AI on economic outcomes (Noy and Zhang, 2023; Brynjolfsson et al., 2025) and economic research (Horton, 2023; Korinek, 2023).

from existing datasets with machine learning. We add to this literature by providing a scalable method for generating qualitative data from human subjects, demonstrating empirically that AI-led interviews are rich enough to support hypothesis generation. We also show how LLMs can assist with the analysis and coding of interview transcripts at scale while maintaining close contact with the raw data.

The remainder of the paper is organized as follows. Section 2 provides details about the AI interviewer and our empirical applications. Section 3 examines the interview process before Section 4 presents our evidence that AI-led interviews can generate novel hypotheses. Section 5 benchmarks AI-led interviews against survey methods and Section 6 examines the internal and predictive validity of interview responses. Finally, Section 7 concludes.

2 Methods and applications

We explain the scalable implementation of AI-conducted interviews in Section 2.1 and describe the design and interview structure of our empirical applications in Section 2.2. Methods for large-scale text analysis of these interview transcripts are presented in Section 2.3.

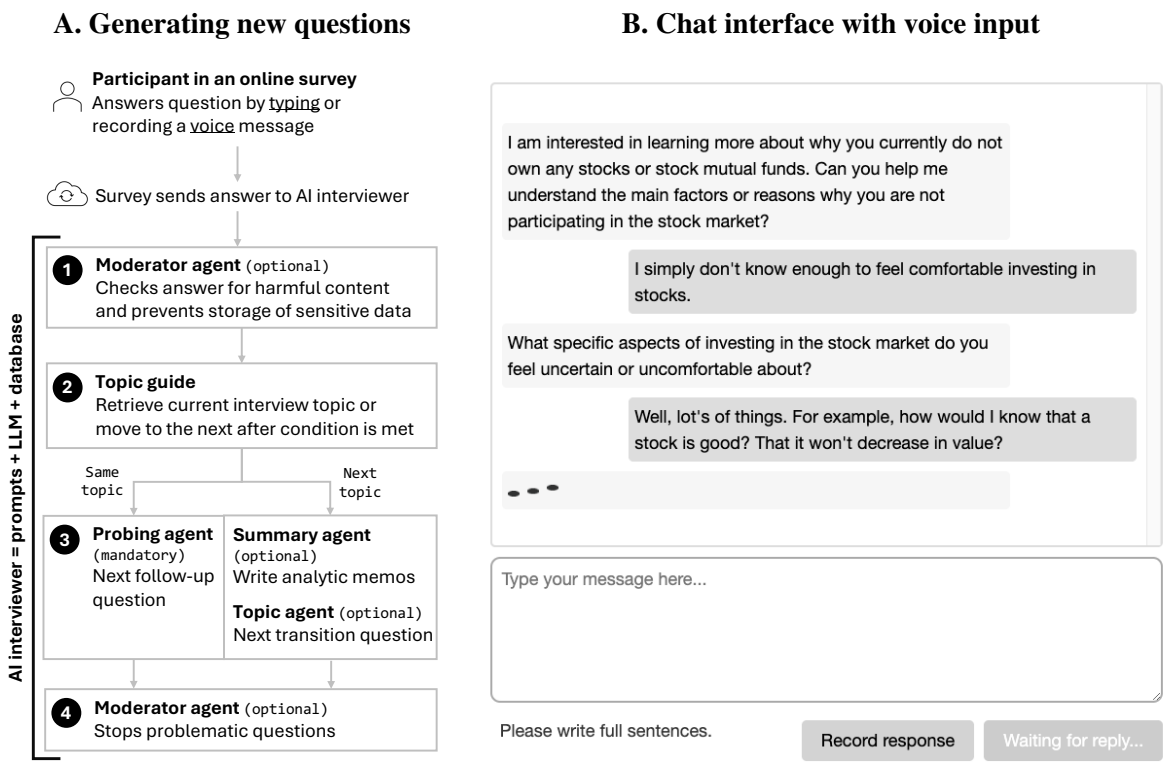
2.1 Conducting interviews at scale with an AI interviewer

Qualitative interviews differ from surveys because participants can express their views in their own words in an unconstrained way, and interviewers have the opportunity to ask *adaptive* follow-up questions—in short: flexible, open-ended interaction. This flexibility offers three advantages over surveys, addressing common limitations of stand-alone open-ended survey questions. First, while responses to stand-alone open-ended survey questions tend to be short and ambiguous, interviews generate detailed conversational records that offer the necessary context for interpretation and classification. Second, interviews comprehensively measure how people reason and interpret the world, and thus do not limit the researcher to participants' immediate, top-of-mind reactions. Third, interviews facilitate the discovery of novel hypotheses as unanticipated themes can emerge since—in contrast to surveys—questions are less predetermined by the researcher's prior assumptions about what is relevant or important.[5]

Design objectives Our goal is to maintain the above benefits of interviews when conducting interviews at scale with an *AI interviewer*. In this paper, we focus on *semi-structured interviews* (Knott et al., 2022), where researchers define an interview topic guide in advance but retain the freedom to adjust their line of questioning within each topic based on interviewees' responses. This structure ensures comparability across interviews while preserving the interview's exploratory strengths (DeLuca et al., 2023). To maintain these benefits at scale, an AI interviewer must meet three core design objectives. First, it must be capable of conducting high-quality

[5] As a detailed review and discussion of interviews and qualitative methods is beyond the scope of this paper, we defer the interested reader to the many excellent textbooks and overview articles that have been written on these subjects (e.g., Creswell, 2009; Mohajan, 2018; Small and Calarco, 2022; Knott et al., 2022).

Figure 1: The AI interviewer and the participant experience



Note: Panel A provides an overview of the workflow of how the AI interviewer generates the next interview question. Agents marked optional can be individually deactivated to turn the multi-agent workflow into a single-agent workflow. In this case, the probing agent handles both topic transitions and the generation of follow-up questions. Panel B provides a screenshot of the 2025 chat interface with the option to either answer in writing or by recording a voice message that is transcribed automatically.

interviews—asking neutral, open-ended questions, but also engaging in adaptive probing that responds meaningfully to what the participant says. This requires reading between the lines, responding to contextual cues, revisiting relevant earlier points, and introducing new topics without disrupting the flow of the conversation. Second, it must be practically reliable: robust to unusual input, while fast and consistent in generating high-quality output. Finally, from an ethics perspective, the AI system must be safe for use with human participants. We next describe how we meet these objectives in practice.

Modular system: Single or multi-agent workflow Our AI interviewer is surprisingly simple and modular, consisting of a database, a set of prompts, and a large language model. At any stage of the interview, the next question is dynamically generated by *prompting* an LLM via an Application Programming Interface (API). We started with OpenAI's `GPT-4` base model in 2023 (without further fine-tuning the model), and switched to `GPT-4o` in 2025, but other frontier or open-source models can be used if preferred.[6] The workflow for generating the

[6]Frontier LLMs can conduct interviews in other languages beyond English, provided the language is well-represented in the training data. For example, our system has been used to conduct interviews in German with high

next question is shown in Panel A of Figure 1. Researchers can switch between a (simpler) single-agent or a multi-agent workflow with more fine-grained control by activating agents indicated as optional. After participants answer a question, the survey software forwards the response to the AI interviewer (a Python application hosted on a server). To generate the next question, up to four distinct tasks are carried out by up to five different *agents*: the probing agent (required), the moderator agent (optional), the summary agent (optional), and the topic agent (optional). Each agent is an LLM with a unique *prompt* of about 150-350 words, i.e., short written instructions describing the desired behavior and output. Together, they collectively function as our AI interviewer. This *decomposed prompting* approach that splits the complex task of interviewing into simpler tasks for different agents is a common strategy to improve LLM performance (Khot et al., 2023). Our modular approach allows researchers to add or subtract agents as needed. For instance, a fully unstructured interview without a topic guide can be accommodated by deactivating the topic agent. For a longer interview, the summary agent can be activated to compile analytical notes on the fly as input for other agents. We discuss how to modify or extend our modular system in Section D.2.

The workflow for generating questions starts with the **moderator agent (optional)** that verifies whether the participant's message includes harmful content, sensitive personal information (e.g., names, credit card details), or attempts to steer the conversation off topic. Flagged messages are not stored and participants are gently nudged to rephrase or decline to answer. The interview stops if messages are flagged too often.[7] After the moderator agent, the next step is to consult the interview's **topic guide**. Researchers can fully customize the interview by defining a set of topics that are to be covered sequentially with a question "budget" per topic, thus giving researchers full control over their length (see Table A.1 for examples). Once this question budget for the current topic is exhausted, the conversation moves to the next interview topic.[8] When transitioning to a new topic, the **summary agent (optional)** starts by updating an analytic summary of the conversation so far for the other agents. The summary agent is designed for very long interviews that require compressing the text input to focus the model's attention. After the summary is updated, the **topic agent (optional)** is responsible for introducing the next interview topic. The agent is instructed to find a smooth and natural transition to bridge what has been discussed with what will be covered next. If no topic transition is needed, the next question is instead generated by the **probing agent** (the only required agent in our setup). The probing agent receives the

respondent satisfaction (Baader et al., 2025). Moreover, they can understand and respond to text with severe spelling errors and non-standard grammar, making them particularly useful for interviewing participants with lower literacy levels.

[7] In practice, fewer than 1% of messages are flagged, and most flagged messages are false positives. Note that concerns about participants disclosing sensitive information, although instructed not to, apply equally to open-ended survey questions that lack the safeguard described above. OpenAI's API neither stores nor uses data for training.

[8] When deactivating the topic agent, the probing agent endogenously determines the number of questions per interview topic depending on how the conversation unfolds (see Section D.2). However, we argue that a *deterministic* workflow might often be preferred: Fixing the number of questions renders interviews more comparable and offers more *control*, avoiding the risk of having some very short or very long interviews. This is practically important when participant recruitment requires announcing the expected study duration in minutes ex-ante.

current interview topic and the previous conversation (or, optionally, the analytic summary) as context for thinking about a probing, clarification, or follow-up question to achieve both depth and breadth of coverage. In a final step, the **moderator agent (optional)** reviews and, if needed, stops potentially problematic questions from being asked as a final safeguard.

Prompts As qualitative methods textbooks are part of their training dataset, frontier large language models already have a strong baseline understanding of qualitative interviews, and thus require only minimal instructions and reminders to activate this context. For each agent, we curate individual prompt templates of no more than 350 words. These prompt templates include best practices for interviewing that we compiled from standard textbooks (Legard et al., 2003; Rubin and Rubin, 2011; Knott et al., 2022; Corbin and Strauss, 2014). The prompts for each agent follow the same basic structure. We first introduce a context where the AI interviewer is informed that it is an "AI proficient in conducting qualitative interviews for academic research" and inform the AI about the general topic and context of the interview. Establishing a role for an LLM ("you are an…") has been shown to increase the performance of LLMs (Kong et al., 2024). After providing the previous conversation, analytic summary and information about the current topic as inputs, we describe the task and guidelines. To make this concrete, we present the core part of the probing agent's prompt below and relegate the other prompts to Section D.1:

> TASK: Your task is to formulate the next probing question for the Current Conversation. The question should align with the Current Interview Topic, helping us to better understand *[insert interview topic]*.
>
> GENERAL GUIDELINES:
> 1. Open-endedness: Always craft open-ended questions ("how", "what", "why") that allow detailed and authentic responses without limiting the interviewee to "yes" or "no" answers.
> 2. Neutrality: Use questions that are unbiased and don't lead the interviewee towards a particular answer. Don't judge or comment on what was said.
> 3. Respect: Approach sensitive and personal topics with care. If the interviewee signals discomfort, respect their boundaries and move on.
> 4. Relevance: Prioritize themes central to *[insert interview topic]*. Don't ask for overly specific examples, details, or experiences that are unlikely to reveal new insights.
> 5. Focus: Generally, avoid recaps. However, if revisiting earlier points, provide a concise reference for context. Ensure your probing question targets only one theme or aspect.
>
> PROBING GUIDELINES:
> 1. Depth: Initial responses are often at a "surface" level (brief, generic, or lacking personal reflection). Follow up on promising themes hinting at depth and alignment with the research objective, exploring the interviewee's reasons, motivations, opinions, and beliefs.
> 2. Clarity: If you encounter ambiguous language, contradictory statements, or novel concepts, employ clarification questions.
> 3. Flexibility: Follow the interviewee's lead, but gently redirect if needed. Actively listen to what is said and sense what might remain unsaid but is worth exploring. Explore nuances when they emerge; if responses are repetitive or remain on the surface, pivot to areas not yet covered in depth.

As qualitative disciplines and scholars strongly differ in their views on how to "best" conduct interviews, the above prompt template focuses on very minimal aspects that most researchers would agree on (e.g., open-ended, non-leading questions, clarifying ambiguities, refraining from judgment, and so on). If the AI interviewer should follow a particular style of interviewing,

researchers can modify our prompt templates in that direction. The above prompt contains no application-specific instructions, making it portable across domains.[9]

Interview interface: Text or voice input Participants interact with the AI interviewer through a chat interface (Panel B of Figure 1) that can be embedded into standard survey software, making it compatible with both online interviews and computer-assisted personal interviews in the field. It resembles common text messaging applications to minimize technological barriers for participants. Participants can respond to questions either by typing directly into the text box or by recording a voice message that is transcribed with OpenAI's Whisper API (without storing the audio message). Participants can thus choose the input method—audio or text—that best allows them to convey their thoughts.[10] As of 2025, the waiting time until the next question is displayed is very short and essentially negligible with a median duration of 1.3 seconds.

Open-source software To facilitate adoption by other researchers, our AI interviewer application is open-source with detailed documentation and setup scripts to run the model either locally or hosted on a server.[11] The local setup is ideal for experimentation and adaptation of the setup to a new research context. For eventual data collections, we provide production-ready code that can be deployed on standard cloud infrastructures (e.g., Amazon AWS for about $1 per month). We provide custom code to integrate the chat interface into standard Qualtrics surveys (by copy-pasting HTML and JavaScript into a "Text/Graphic" question). Integrating the AI-led interview directly into the survey environment reduces respondent attrition by eliminating the need to navigate away from the survey and increases flexibility by allowing customization of prompts based on previous survey responses.

2.2 Applications: Design and interview structure

The remainder of the paper focuses on a comprehensive examination of AI-led interviews, where we evaluate the following four key questions. First, do AI-led interviews keep respondents engaged and satisfied throughout the interview? Second, does the AI interviewer adhere to its instructions, securing alignment with best-practice interview standards? Third, do the interviews produce high-quality data that provide rich insights into economic behavior and economic reasoning? And fourth, do AI-led interviews add substantial value beyond surveys with open-ended questions? To investigate these questions, we conduct about 800 AI-led interviews in two complementary domains for understanding (i) individual economic behavior and (ii) economic policy views. Together, these applications cover a large share of experimental and survey-based research in economics. We describe the study design and interview structure of our

[9]The total LLM usage costs for designing the AI interviewer and developing our prompts in 2023 were negligible, amounting to less than $50 in 2024 prices.

[10]Evidence from randomized controlled trials suggests that spoken responses yield content of comparable richness to written responses (Namey et al., 2020), a finding that we replicate in our second application where the voice input feature was added to our setup (see Section 5.3).

[11]The full code and documentation are available here: `https://github.com/fchop/interviews`

concrete applications below, and introduce additional data collections throughout the paper when relevant. Table A.2 provides an overview of all studies and Section F contains the experimental instructions.

2.2.1 Economic behavior: Stock market non-participation

To investigate whether AI-led interviews generate valuable data to study economic behavior, we selected the stock market non-participation puzzle as our first application and ask non-investors why they do not participate in equity markets (Guiso and Sodini, 2013). This puzzle is of great interest to economists and policymakers, and the substantial literature on the puzzle sets a high bar for AI-led interviews to potentially generate a novel hypothesis about the underlying drivers.

Sample Our analysis sample consists of 381 interviews with US Americans conducted between August 23 and September 5, 2023.[12] Participants were recruited from Prolific, a survey platform commonly used in economic research that is associated with high data quality (Peer et al., 2022). We use a screener survey with standard requirements to recruit the relevant subpopulation of non-investors with the financial means to participate in equity markets (see Section F.1.1 for details). To ensure a high quality of responses as required for interviews (Rubin and Rubin, 2011), the screener survey includes a standard attention check and additionally screens out participants who demonstrate an unwillingness to engage with an open-ended survey question. We exclude 14 participants flagged during a human review of all interview transcripts as having used AI tools. Table A.3 provides summary statistics for our sample, which deviates from the general population as we target non-investors. Our sample is close to the US population in terms of age, but has a higher share of women and participants with a college degree, as is common for online panels (Armantier et al., 2017). The average household income is $72,000, with 60% reporting liquid savings equivalent to at least two months of income. Reassuringly, net financial assets are about $51,000, indicating that most participants have the means to invest in stocks.

Design and interview structure All interviews were conducted in writing (without our voice input feature) as part of an online survey with a median completion time of 40 minutes and a fixed compensation of $12. The survey starts with a detailed consent form disclosing that the study includes a chat-based interview with an AI chatbot to allow participants to select out of the study if they feel uncomfortable with either an interview or engaging with a machine (a selection decision we will study in Section 3). We also inform them that their responses will be processed via OpenAI's API, but that their data will neither be used for model training nor permanently stored by OpenAI. Before the interview, we elicit background characteristics using standard survey questions and allow participants to familiarize themselves with the chat interface. After the interview, we elicit detailed feedback on the interview experience and include additional survey questions on factors correlated with stock market non-participation.

[12] The field time is thus substantially shorter compared to sequentially conducted interviews with a human interviewer, and only required several days due to OpenAI's API rate limit in 2023.

The interview itself is semi-structured with four conversational topics, with the verbatim instructions provided to the AI interviewer for each topic shown in Table A.1. For this data collection, the AI interviewer consisted of a summary agent, a topic agent, and a probing agent.[13] The conversation always starts with a broad question that signals interest in the participant's personal reasons for non-participation: "I am interested in learning more about why you currently do not own any stocks or stock mutual funds. Can you help me understand the main factors or reasons why you are not participating in the stock market?"[14] After five follow-up questions, the conversation moves to the next topic that instructs the AI interviewer to explore the "perceived barriers or challenges preventing [the respondent] from participating in the stock market." We chose this topic to ensure that the interview covers other relevant reasons for non-participation, as it naturally "reorients" a conversation stuck in a single line of inquiry. The third topic asks participants to walk the interviewer through a hypothetical scenario in which they own stocks (three questions), employing the commonly used strategy of counterfactual reasoning to uncover participants' reasoning about what it means to own stocks. The fourth and final topic revolves around "conditions or changes needed for the interviewee to consider investing in the stock market." Following common practices (Rubin and Rubin, 2011), the interview concludes with two questions on whether participants would like to share anything else and asking them to reflect on the conversation and again identify their main reason for non-participation. After the interview, we present the conversation summary generated by the summary agent and ask participants whether it accurately represents their views expressed in the interview.

2.2.2 Economic policy views: US import tariffs

To examine the value of AI-led interviews for studying economic policy views, we interview US Americans about their views on the US import tariffs imposed by President Trump in 2025. Tariffs are a policy issue on which laypeople's accounts are likely to diverge from standard economic theory, rendering open-ended qualitative data especially valuable (Mutz, 2021).

Sample In May and June 2025, we conducted 385 online interviews with US Americans recruited from Prolific with a screener survey that elicits baseline support for tariffs and includes similar quality requirements as described above.[15] Table A.4 provides summary statistics. 30%

[13] At the time of the data collection, the relatively short context window of GPT-4 made a summary agent strictly necessary. While current frontier models have longer context windows, a summary agent might still be useful to guide the model's attention in very long interviews. Furthermore, the final conversation summary can be shown to the interviewee, allowing them to rate whether the interview accurately captured their views.

[14] While interviews usually start by building rapport, we deliberately chose this question because it resembles the type of open-ended survey questions economists use to gather qualitative data. We later contrast the themes revealed in participants' responses to this question with the themes a researcher would uncover based on a full interview.

[15] We take comprehensive measures to mitigate growing concerns about bots and LLM use in online surveys. To prevent the use of AI tools, we programmatically block participants from copying or pasting text, rendering tools like ChatGPT useless. In addition to Captcha/reCaptcha methods, we track—among others—the ratio of text produced to unique keystroke events to identify bots in our screener survey that "produce" text without hitting a physical keyboard. A final human review of the interview transcripts did not find any evidence of remaining bot activity.

of our sample support and 70% oppose the 2025 US import tariffs.

Design and interview structure Interviews are again conducted as part of an online survey. We measure background characteristics before the interview and participant feedback, as well as an additional survey module eliciting views about tariffs using structured questions, after the interview. For this data collection, we used the moderator, probing, and topic agents. Moreover, a key difference from our first application is that participants could now answer interview questions either by typing their response or by recording a voice message that we automatically transcribe. The interview itself is again semi-structured, following the topic guide outlined in Table A.1 with the opening question: "First of all, what are the main reasons you support or oppose the new 10 percent tariff on imports?" After five follow-up questions, the AI interviewer uses a counter-perspective prompt, asking whether there are any positive (negative) aspects of tariffs if the participant previously discussed primarily negative (positive) aspects. The third part asks participants to explain to someone unfamiliar with tariffs how they work, while the fourth part explores participants' beliefs about the consequences in a scenario where the US raises tariffs to 30%. Finally, the AI interviewer is tasked with reviewing the conversation and following up on promising but unexplored aspects before concluding the interview.

2.3 Methods for analyzing interviews

This subsection explains the general methods we use to (1) qualitatively analyze transcripts and inductively construct a coding manual, (2) conduct human coding of interviews, and (3) validate AI coding of interview transcripts against a human benchmark. All methods are designed to scale to a text corpus of millions of words while retaining a "human in the loop." Similar to how applied researchers are expected to understand their raw data and setting, it seems prudent for those working with interviews to stay close to the source text to avoid losing nuance and context during interpretative aggregation (see Nelson, 2020, for the full argument).

2.3.1 Discovering themes and inductively building a coding manual

We use inductive thematic analysis to analyze our transcripts, i.e., we aim to discover notable themes solely from what participants say rather than imposing what we as researchers believe should be present in the data (Knott et al., 2022). This approach is deliberately agnostic and aids the discovery of novel ideas in contrast to deductive methods that derive codes from theory (Glaser and Strauss, 2017). A key output of this process is a collection of *codes* that each represent a notable, recurring theme (e.g., concerns about the costs of owning stocks, or concerns about tariff-induced inflation). The codes jointly define a so-called *coding manual*, which can be used for labeling responses. To arrive at a scalable method, we borrow techniques from computational sociology (Nelson, 2020). First, we independently read a random sample of 50 interview transcripts to obtain a robust baseline understanding of the context and the language participants use. In a second step, we transform interview transcripts into a text corpus where

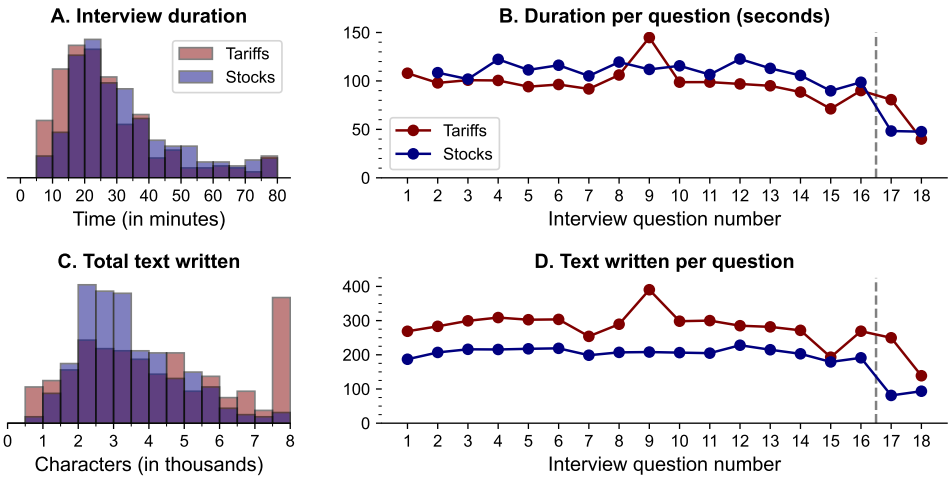
each document is a question-answer pair from an interview (about 13,000 pairs per application). We then use BERTopic, an unsupervised embedding-based clustering algorithm, to sort the data into clusters of semantically similar text (Grootendorst, 2022). The motivation for this step is that it leverages the pattern recognition ability of machine learning to increase the chance and speed of discovering a notable theme in the data (Ludwig and Mullainathan, 2024). In contrast to bag-of-words or word list methods, BERTopic constructs more semantically coherent topic clusters than non-embedding-based methods that ignore the semantic relationship of words. In a third step, we independently and sequentially review each cluster, focusing on the model's top-ranked responses. For notable themes, we write a short synopsis, code definition, and compile representative example quotes. At the end, we compare notes, consolidate our list of codes, and ensure a shared conceptual understanding of codes. The above process, while described linearly, was iterative in practice.

2.3.2 Hand-coding and AI-coding interviews

Human hand-coding As a ground truth for validating the quality of AI coding, a team of research assistants trained in qualitative interview methods hand-coded over 18,000 questions and responses from our interview transcripts. Appendix Section E provides details about the training, calibration of annotators, and coding practices. The interrater agreement between research assistants is 83%, indicating a high quality of the hand-coded data. We annotate interviews twice, once using the coding manual derived from the inductive thematic analysis, and another time using a second coding manual focused on rating interviews based on explicit quality criteria for interviewing. We fully hand-coded all stock market interviews, while a random half of all tariff interviews are hand-coded for the analyses presented in Section 3.

Validated AI coding Hand-coding is the gold standard for annotating qualitative text data, but the time and labor costs render it ineffective for working with interviews at scale. To address this, we demonstrate that researchers can use LLMs to automate the hand-coding process. Of course, it remains crucial to validate the performance of AI coding against a human benchmark to guard against hallucination and systematic model biases (Carlson and Dell, 2025). For AI coding, we use OpenAI's `GPT-4o` model. Section E.2 contains the prompt, which includes general coding instructions, such as avoiding implicit assumptions, a placeholder for the definition of the code, and a list of example responses. To increase accuracy, we ask the model to supplement its decision with a justification and to cite evidence from the transcript. Using data from the stock market interviews, we find that AI coding closely aligns with the human-coded ground truth, as shown in Figure B.1. The global interrater agreement between AI and human coders across all codes is 86%, and thus as high as the human–human agreement in our setting. At the code level, the agreement ranges from 81% at the 25th percentile to 92% at the 75th percentile, with `GPT-4o` on average being 6.4 percentage points more likely to assign a code to an interview than a human. Taken together, this suggests that human-validated AI coding can serve as a

Figure 2: Text production and thinking time are high and stable over the course of the interview



Note: Data from the stock market and tariff interview studies. Panel A: Histogram of the interview duration in 5-minute bins and censored at 80 minutes. Panel B: Average time to answer the kth question in the interview in seconds. For the stock market interviews, the response time to the first question could not be recorded yet. Panel C: Histogram of the total text written by participants in thousands of characters and censored at 8,000 characters. For the tariff interviews, the mass at 8,000+ characters is driven by participants who used the voice-to-text input method. Panel D: Average text written in response to the kth question in the interview in characters. Dashed lines in Panels C and D indicate that questions 17 and 18 are the closing questions of the interviews.

cost-effective substitute for labeling interview transcripts.

3 Behavioral and perceptual markers of interview quality

Before turning to the thematic content of our interviews, we first examine the quality of the interview process itself. This step is essential: because AI-led interviews are conducted without direct researcher involvement, we cannot rely on anecdotal observation to assess what the AI interviewer is doing or how participants are responding. Instead, we assemble a set of behavioral and perceptual markers that should be indicative of high interview quality. These include the time and effort participants invest (Section 3.1), the quality of the questions asked by the AI interviewer as judged by trained human annotators (Section 3.2), participants' subjective evaluations of the interview (Section 3.3), and an experiment measuring demand for AI-led interviews (Section 3.4). What unites these exercises is that they do not depend on analyzing the content of participants' responses. Instead, they offer an external lens on the interaction itself. These process-level metrics are a necessary foundation before turning to the content-based analyses in Section 4, 5, and 6. They speak to the quality of the interview as an interaction, i.e., whether it fosters engagement, adheres to methodological best practices, and creates the conditions under which meaningful qualitative data can be obtained.

3.1 Interview duration and participants' effort

We first present descriptive evidence that participants take the interviews seriously and exert high effort answering the questions. As shown in Panel A of Figure 2, many participants spend considerable time completing the interview. The stock market interviews take about 33 minutes on average to complete, and participants produce 654 words (3,487 characters) of text during this time (Panel C). With 29 words per minute, the text output is comparable to the benchmark of about 20 words per minute obtained in chat-based interviews with human interviewers (Namey et al., 2020). The tariff interviews are slightly shorter with a median duration of about 25 minutes, but text production exhibits a significant right tail, which is driven by participants who used the voice-to-text input feature. We return to this in Section 5.3.

Over 90% of participants complete the full interview, which is striking given that participation in the interview was voluntary and participants knew payments were not conditional on participation. In fact, respondents could decide to proceed in the survey and skip the interview after a five-minute delay. The very low attrition rate is encouraging given concerns that survey respondents might not exert effort without meaningful incentives, especially for open-ended questions that require more effort from participants compared to closed-ended questions (Haaland et al., 2023). Notably, effort as measured by text output or thinking time does not decline over the course of the interview. Participants consistently take about 100–120 seconds to answer interview questions (Panel B), writing about 200–300 characters on each question, with the only understandable drop for the closing questions that signal the end of the interview (Panel D).

3.2 Assessing the behavior of the AI interviewer

We next open the black box of what happens during interviews and provide descriptives to answer the relevant question: What is the AI interviewer doing? To do so, we follow best practices and use human judgments as the gold standard for evaluation (Dell, 2025).[16] As described in Section 2.3.2, we systematically hand-coded over 8,000 question-answer pairs on 10 markers of interview quality. These markers capture both local properties of questions (e.g., using open-ended rather than closed-ended wording) and contextual properties that depend on the previous conversation (e.g., whether a question is novel or rephrases an earlier question). Details about the coding manual are shown in Appendix Section E.

Table 1 presents the results. Reassuringly, we find that the AI interviewer consistently adheres to core methodological guidelines for interviewing across both applications. First, about 95% of questions in the stock market interviews are coded as being (i) open-ended rather than closed-ended, (ii) relevant to the interview's objective, and (iii) non-leading, indicating that the

[16] It is a common problem with large language models that they often fail to follow explicit instructions (Zhou et al., 2023). Furthermore, Kamoi et al. (2024) find that even frontier models such as GPT-4 and Claude 3 detect errors "at very low recall" and are "much worse than humans" at error detection, making it essential to use trained *human* evaluators to evaluate the quality of the interviews and LLM alignment with instructions.

Table 1: Characterizing the behavior of the AI interviewer

	(1) **Economic behavior** (Stock market)	(2) **Policy views** (Tariffs)
Local properties		
Open-ended question asked	94.8%	99.8%
Question relevant to the interview objective	95.0%	97.9%
Neutral and non-leading question	95.7%	99.6%
Contextual properties		
Novel, non-repeated questions	82.7%	92.2%
Follow-up about the participant's last answer	77.6%	37.5%
Follow-up about something mentioned earlier	15.4%	13.9%
Ethics and safety		
Signs of high conversational engagement	79.6%	85.2%
Annoyance with the interviewer	2.5%	0.7%
Any potential signs of discomfort	0.2%	0.1%
LLM hallucinated	0.1%	0.4%

Note: Data from the stock market and the tariff interview studies (see Table A.2). Summary statistics derived from hand-coding by a team of trained research assistants with interviewing experience. All percentages are computed at the question-answer pair level, excluding the predetermined closing questions. Local properties of questions are assigned by judging the question in isolation, while contextual properties require knowledge of the previous conversation.

AI does not ask biased questions. For the tariff interviews—which were conducted with a more advanced LLM—these shares are close to 100%, confirming our expectations that interview quality will naturally improve over time as foundational models naturally mature.

Second, turning to contextual properties, we find that 83% (92%) of stock market (tariff) questions are unique and thus not a rephrasing of an earlier question, which research assistants were instructed to conservatively classify as "redundant"—although repetition can be useful for revisiting earlier points, clarifying the meaning of the question, or prompting a participant to tell us more. These markers of question quality capture core principles for interviewing that we *explicitly* described in our prompts for the AI interviewer, and thus speak to the AI interviewer's instructional alignment.

An important skill for conducting interviews is understanding contextual cues and establishing a *dialogue* where questions dynamically respond to participants' answers in a meaningful way. For the stock market interviews, we find that 76% of all questions explicitly refer to something that the participant said in their most recent response; for example, by asking a follow-up question about a newly raised point. This share is lower for the tariff interviews, where we made a minor prompt change to reduce the tendency to paraphrase the participant to make questions shorter. Moreover, 14–15% of interview questions revisit a key theme raised by the participant during earlier parts of the conversation.

Third, we assess markers of participants' well-being and safety, arguably key objectives for interviewing. Research assistants hand-code responses for any potential signs of discomfort

or annoyance with the interviewer. Reassuringly, signs of discomfort are extremely infrequent. Only 2.5% of responses indicate annoyance in our stock market interviews. These instances were typically due to respondents feeling they had already answered a question, which explains why the share is only 0.7% for the tariff interviews that include fewer repeated questions. Similarly, we find that the share of responses rated as displaying signs of high conversational engagement—which focused on whether participants make an effort to actively participate in the conversation rather than the amount of text produced—ranges from 80% to 85% across applications.

We also address concerns about model hallucination (Zhang et al., 2023), which cannot be credibly addressed without a human coder carefully analyzing each interview transcript. We detect only five (seven) instances of questions that include incorrect references to prior responses in our stock market (tariff) interviews, suggesting that hallucination is effectively absent in our contexts.[17]

Finally, we provide evidence that the AI interviewer also displays an understanding of when clarification questions are needed. Figure B.2 shows that the probability of asking a clarification question increases sharply when interviewees' responses are hand-coded as requiring further clarification. Clarification questions are rare (2.8%) when humans perceive no need for further clarification, but the share increases to 31% and 51%, respectively, when clarification is deemed "somewhat relevant" and "very relevant." This result suggests that the AI interviewer is able to recognize ambiguous responses and follow up with a clarification question.

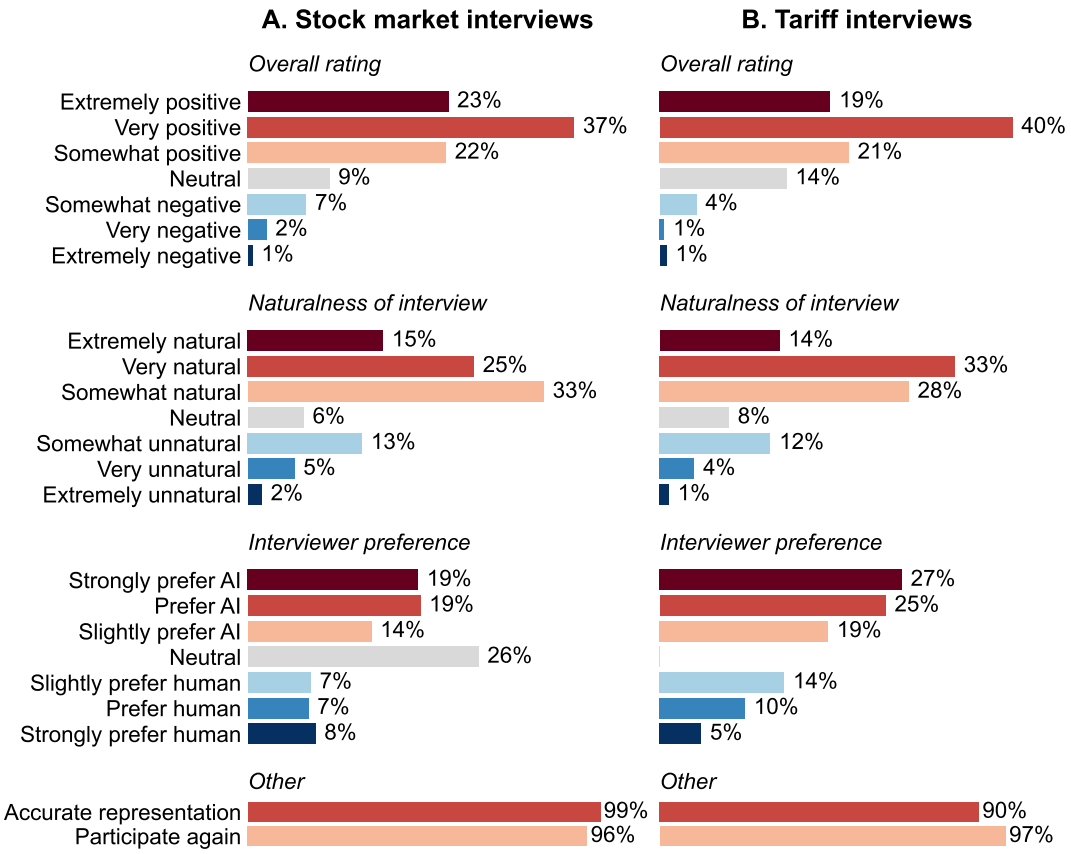
Taken together, these findings provide a direct lens into AI interviewer behavior, indicating that the AI interviewer consistently follows key methodological instructions for interviewing and displays high contextual awareness during conversations, while ensuring participant safety.

3.3 Participant perceptions of the interview

After the interview, we elicited participants' perceptions and evaluations of the interview to obtain direct evidence on whether the interviews are perceived as a positive and natural experience by the interviewees. Focusing first on the overall assessment, Figure 3 shows that participants in both applications provide overwhelmingly positive reviews, with about 60% rating the interview as "very" or "extremely positive" across contexts. When asked about the naturalness of the conversation, most participants rate the conversation as feeling natural. We also asked participants whether they would prefer a human or an AI interviewer in a future interview on a comparable topic. For our stock market interviews, 53% indicated a preference for an AI interviewer compared to only 21% who prefer a human interviewer. For the tariff interviews, we removed the neutral option, which increases the share of participants who prefer an AI interviewer to 71%. This suggests that the *experience* of participating in an AI-led interview must be sufficiently

[17] We coded this conservatively. Most cases are minor hallucinations where the interviewer made an inference error based on the previous conversation, such as concluding that the participant is interested in learning more about the stock market when this was not explicitly said.

Figure 3: Participant evaluations and perceptions of the AI-led interview



Note: Panel A uses data from the stock market interviews study ($n = 381$), while Panel B uses data from the tariff interview study ($n = 385$; see Table A.2). "Overall rating" shows participants' overall evaluation of the interview. "Naturalness of interview" is participants' assessment of the naturalness of the conversation. "Interviewer preference" is the response to the question whether they would prefer a human or an AI interviewer in a future interview. Note that the "neutral" response option was removed for the tariff interview study. "Accurate representation" is the share of participants approving the summary of the interview as accurate in Panel A, and shows the share saying that the interview somewhat or very accurately captures how they think about tariffs in Panel B. "Participate again" is the share that would like to participate in another AI-led interview.

positive. Indeed, almost all of our participants (96–97%) indicate that they would like to participate in another AI-led interview.

Finally, we show that AI-led interviews are able to accurately capture participants' perspectives. After the stock market interviews, we asked participants to approve a summary of why they are not participating in the stock market: 99% approve the summary as accurately capturing their case. After the tariff interviews, we ask participants to rate whether the interview "accurately captures how [they] personally think about the topic of tariffs" on a 5-point Likert scale and obtain similar results. Figure B.3 provides additional direct evidence on how participants perceive the interview questions in the tariff interview, showing that questions are rated as (i) open-ended, (ii) neutral, and that participants (iii) express feeling comfortable during the interview.

3.4 Selection experiment: Revealed demand for AI-led interviews

Design The above findings suggest that people display a high willingness to engage with an AI interviewer and might even prefer AI interviews over human interviewers. To mitigate potential concerns about social desirability bias or positive selection into our AI interview samples, we ran a correspondence study with 500 US Americans recruited from Prolific in November 2023 using equivalent recruitment criteria (see Section F.1.5 for instructions). Table A.3 provides summary statistics for the sample. In the survey, we inform participants that they might be eligible to participate in three additional studies. We describe each study on a separate survey page in randomized order. Each study is described analogously to how we described the stock market interview study on Prolific. However, we exogenously vary whether the study is described as (i) "a 40-minute survey," (ii) "a 40-minute study with a text-based interview with an AI," or (iii) "a 40-minute study with a text-based interview with a human." On the page where the study is described, participants indicate whether they would like to be invited to the study if we decide to run it.

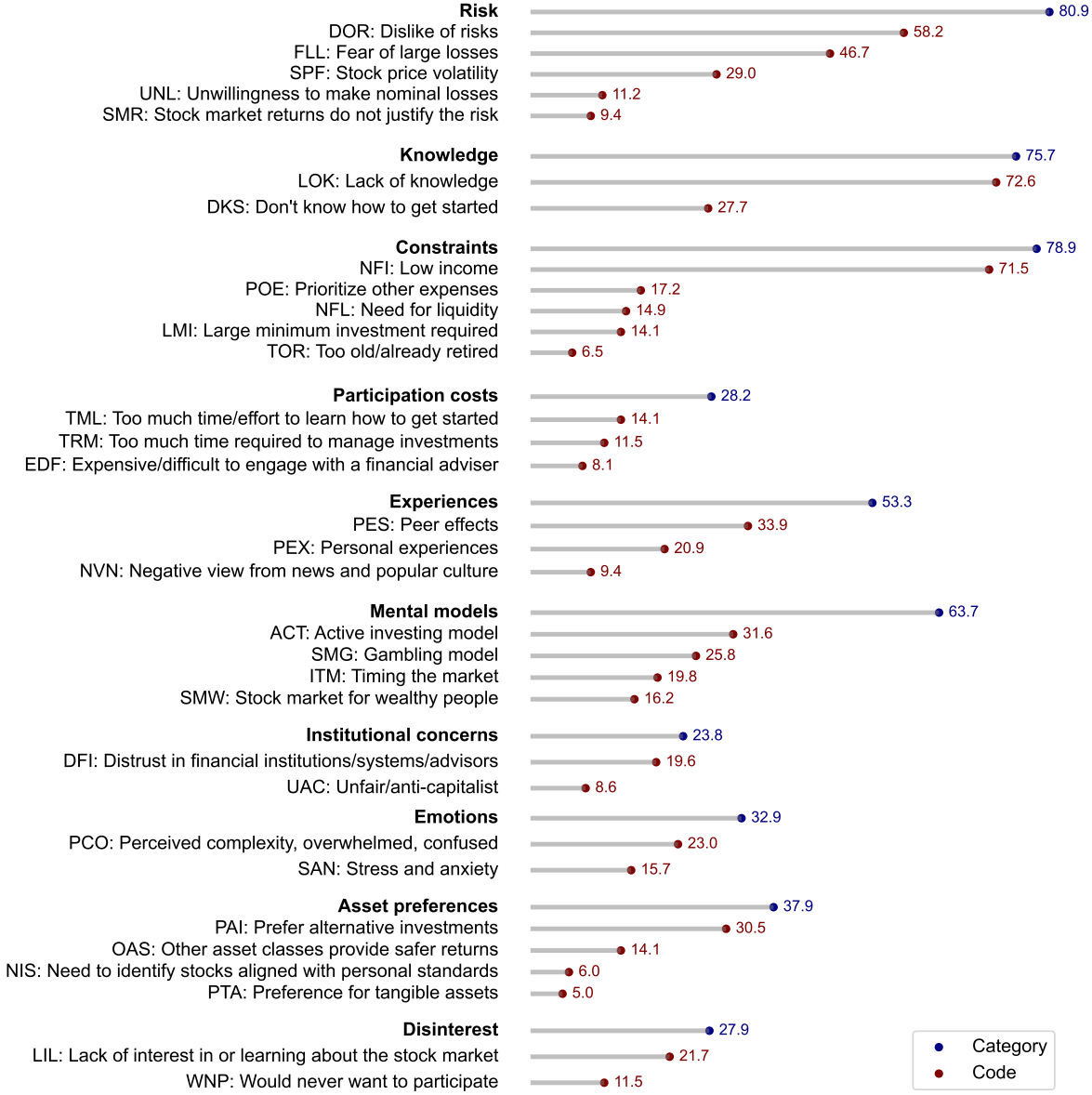
Results As a clean comparison without spillover, we focus on the first study described to participants. Figure B.4 shows that demand for study participation is identical for a regular survey (89%) and a survey with an AI-led interview (90%). If anything, demand is 6.6 percentage points *lower* for a survey with a human-led interview ($p = 0.078$). Moreover, exploiting within-subject variation, we find no evidence of differential selection into surveys compared to AI interviews (as shown in Table A.5). We thus replicate a weak preference for AI-led compared to human-led interviews also from an ex ante perspective prior to participation in an AI-led interview.

4 Qualitative findings and hypothesis generation

Having established that AI-led interviews maintain respondent effort and satisfaction while also adhering to best-practice interview standards, we now turn to whether they also produce rich, high-quality interview data that can be used for hypothesis generation—the hallmark of qualitative research. We present our main *qualitative* findings from the stock market interviews in Section 4.1 and then present an out-of-sample validation of our derived hypothesis in Section 4.2. In Section 4.3, we present key findings from the import tariff interviews. We return to these findings in Section 5 to show that many relevant themes would not have been present in responses to standard open-ended survey questions.

As our primary goal is to evaluate AI-led interviews as a new method for data collection, the qualitative results below are not intended as a complete account of all themes present in the interviews. Rather than focusing on the most frequently mentioned themes, we highlight those that reveal interesting perspectives relative to previous work or that illustrate the added value of AI-led interviews over open-ended survey questions (see Section 5). In each application, we

Figure 4: Prevalence of discovered codes in the stock market interviews



Note: This figure presents the frequency of codes discovered during our qualitative analysis of the stock market interviews (see Table A.2 for an overview of studies). Codes derived from hand-coding. For each code, we present the share of participants to which this code has been assigned at least once. For categories, we create a dummy equal to 1 if at least one code from the category was assigned to the interview, and zero otherwise.

present findings narratively and illustrate them with representative quotes drawn from participant responses. To remain close to the conventions of qualitative research, we follow Bergman et al. (2024) and focus on a rich description of the key mechanism with illustrative quotes from the interviewees. We refer the reader to the relevant figures in each section for code frequencies.

4.1 Main qualitative findings from the stock market interviews

Our interviews uncover considerable heterogeneity in the perceived barriers to stock market participation, confirming that non-participation is a multi-layered and multidimensional phenomenon. Our evidence supports leading explanations of non-participation, such as low financial literacy, risk-related concerns such as fear of losses, peer effects, participation costs, or income constraints (Guiso and Sodini, 2013). As the focus of our paper is methodological, we focus on three main qualitative patterns and present additional qualitative findings in Appendix Section C.2. Figure 4 presents code frequencies.

4.1.1 Mechanism 1: Participation costs and the mental model of active investing

When asked about their reasons for not participating in the stock market, a recurring theme is that respondents describe concerns about the time commitments and the work involved in owning stocks. As one respondent puts it:

> *Managing investments in the stock market feels like it could almost be a part-time job given the level of attention it demands. My current job is already time-consuming and mentally exhausting. Adding the task of researching companies, keeping up with earnings reports, and tracking market trends would be another layer of complexity and stress.*

Many respondents perceive high recurring participation costs from owning stocks, due to the need for (i) research and (ii) monitoring. First, they believe that research is an essential task of an investor: "I know you are supposed to do research on stocks that you would like to invest in, for example if there was a recent oil spill, it would not be good time to buy stock in oil companies." This research has to integrate a broad array of public information, such as historical price data, financial accounts, earnings announcements, and analyst forecasts. As a result, respondents understand that "it takes time to research and understand a company's product and how it fits into the marketplace" and anticipate "spending a couple of hours a day or maybe several hours 3–4 days a week online reading about market trends, emerging technologies or medicines, and companies on the uprise as well as those in decline."

Yet, the perceived job requirements of an active investor continue beyond the point when the initial research is concluded and promising stocks have been identified, because one would "have to constantly monitor the current market condition and make sure that the stock [you] invested in would not suddenly crash." Continuous monitoring is important because stock prices change "all the time, every day" and "if you sell [stocks] at a bad time you wont get as much for the stock than if you sold them at a good time... and waiting for the peak time to sell seems best." Many respondents thus seem to believe that making a return on one's investments requires careful timing of the stock market, which only adds to the perceived costs of participation.

Collectively, these perceptions characterize what we refer to as the *active investing* model of stock market participation. This model is grounded in three fundamental beliefs. First,

respondents believe that market inefficiencies allow investors to predict future stock prices through research and analysis. Second, they perceive stock prices as highly volatile, making continuous monitoring and quick decision-making important to avoid incurring large losses before prices adjust. Third, there is a shared view that investors make returns from short-term trades. These beliefs contribute to a view that participating in the stock market is a demanding, high-maintenance activity, which ultimately deters many respondents from investing.[18]

4.1.2 Mechanism 2: Knowledge barriers

Another prevalent theme among respondents is a concern that they lack fundamental knowledge about the stock market. Many respondents expressed uncertainty about the basic logistics of investing, such as how to get started, how to buy and sell stocks, or create an investment account. As one respondent stated: "I don't know where to view the stock market, how people make money from stocks, and/or how they work. I don't know how to invest, or what to start with when investing." This perceived lack of knowledge extends beyond logistical aspects and into the complexity of the stock market itself:

> *To be honest, all of it sounds complicated. I barely know what any of the terms in the market mean. Like, I'm not familiar with dividends, mutual funds, etc. Aside from not having knowledge of the terms, I also don't know what makes something a good investment vs a bad one. I wouldn't even know where to start.*

Many respondents point to the confusing jargon and technical concepts used in finance. This perceived need to grasp many complicated concepts before even beginning to invest leads many respondents to conclude that acquiring the necessary knowledge would be a time-consuming endeavor with uncertain rewards. As one respondent puts it, "it seems like there is a lot to learn, and that is a little intimidating." Moreover, respondents voiced a deeper concern in the form of a belief that successful investing requires specialized knowledge and skills that they do not possess. Specifically, many respondents explain that successful investing requires an ability to predict stock prices, distinguish "good" from "bad" stocks, and effectively time stock trades ahead of everyone else. As one respondent explains:

> *I feel like I don't know which stocks are 'good' to purchase, or when a good time to purchase would be. Because of this, I feel like I would be flying blind and making decisions entirely at random, as compared to a more seasoned stock market veteran.*

This perceived need for experience and expert knowledge both underscores the cost of stock market participation and reinforces the conclusion that without such knowledge, one is likely to incur losses. Another respondent articulated a similar concern, saying that they are "not quite

[18] Duraj et al. (2024) independently discovers a similar active investing model after conducting 25 human-led qualitative interviews in Germany, providing further robustness to the finding from a different context. Furthermore, since their working paper was made public a year after our findings (Chopra and Haaland, 2023), it could not be the case that our AI interviewer "discovered" this hypothesis because it was part of the training data.

sure how to read stocks and what patterns to look for," and that "for this reason, I feel like if I were to try and invest in stocks without this knowledge I might lose money." This combination of the perceived logistical complexity of investing, a belief in the necessity of extensive knowledge and experience, and the fear of losing money precisely because they lack those skills contribute to a perception of substantial entry barriers to investing.

4.1.3 Mechanism 3: Risk perceptions

A common theme that emerges when discussing the stock market are strong concerns about risks. For instance, many respondents express a deep-seated fear of losing a substantial part, or even all, of their "hard-earned" money on the stock market. Unlike risk-free assets where returns are guaranteed, respondents worry that they could end up with less money than they started with:

> *If I would lose any amount of money, no matter how large or small, it would make me sick to think about. I work so hard for everything I have, that knowing it could all be gone with the stock market failing would make me incredibly anxious.*

A general unwillingness to make nominal losses, no matter how small, is frequently mentioned among participants, who are sometimes explicit that they would need a *guarantee* against making any losses to consider investing. Furthermore, the concern that an investment could "instantly lose all its value" reflects the belief that such catastrophic losses are not only possible, but likely, and could occur at any moment. The fear of catastrophic events leads many to conclude that the risks of investing are too large, and that owning stocks would therefore expose them to constant feelings of stress and anxiety because "the market changes everyday."

The metaphor of gambling frequently surfaces in respondents' accounts of the risks and returns of investing. This comparison is not merely metaphorical but reflects the mental model that the stock market operates with the same uncertainty and lack of control as a game of chance. For these respondents, the stock market is "like a lottery, where you can either win big or lose everything" and it is "all a roulette of which stock will fail and which will win." Investing in stocks is seen as being as risky and unpredictable as gambling, where the odds are perceived to be stacked against the average investor, supporting the view that the stock market is something for "when you have discretionary income to burn. . . not something you can rely on to build wealth."

It is important to note that not all respondents see the stock market purely as a gamble. There is also a belief among respondents that the risks can be mitigated through knowledge and experience. As one respondent noted, investing in the stock market is risky "unless you know what you are doing." Another respondent, who expresses a common concern about going "completely bankrupt," puts it like this:

> *One of my biggest fears is going completely bankrupt since my stocks may fail. I also fear I may not know anything about the company I invest in and something huge happens where they close. I want to feel better about my experience and have that knowledge and funds.*

These ideas tie back to previous themes, where respondents expressed that acquiring the necessary knowledge and skills could transform the stock market from a high-stakes gamble into a more manageable risk, implying that a seasoned investor faces a different risk-return trade-off. Yet, this knowledge comes at a considerable cost both in terms of time spent and, more importantly, the financial losses incurred during the costly learning process, which many feel they could not afford. As one respondent summarizes: "You could put money to buy stocks that end up losing value rather than gaining value and end up in a situation where you lost both time and money for participating." This explains why many respondents prefer risk-free assets where neither time nor expertise are required to receive a guaranteed return on their investments.

4.2 Out-of-sample validation of the generated hypothesis

A key finding from our qualitative analysis of the stock market interviews is that many participants subscribe to an *active investing* mental model that makes stock market participation seem like a very substantial commitment of time and effort. If this mental model indeed acts as a barrier to stock market participation, it should be more prevalent among non-investors than among stock owners. To test this hypothesis, we conduct a large-scale survey in which we measure the prevalence of this mental model with tailored closed-end survey questions. A key benefit of this exercise is that it allows an out-of-sample replication of our qualitative findings which also addresses concerns about "overfitting" whereby spurious patterns in the interview corpus lead to chance discoveries that do not generalize.

Design For the out-of-sample validation survey, we recruited a balanced sample of 1,000 US investors and non-investors from Prolific in May 2024. Table A.3 presents summary statistics and Appendix Section F.1.6 contains the key instructions. We elicit stock market participation and provide a definition of the term. As a direct measure of the *active investing* mental model, we use agreement with the statement below, measured on a 5-point Likert scale:

> "If you want to make any money in the stock market, you have to constantly monitor market conditions, keep up-to-date with all the news, and do a lot of research on the stocks that you plan to invest in." *(Strongly disagree to strongly agree)*

In addition, we elicit other hypothesized facets of the *active investing* mental model. Specifically, we elicit participants' beliefs about how often it would be "necessary to monitor market conditions and check one's portfolio" on a 7-point scale from "less than once a year" to "multiple times a day," their belief about the time per week required to manage one's portfolio, whether it is important to react quickly to daily changes in market conditions and whether it is important to "actively pick a few stocks." We also elicit the types of *public* information that respondents believe one would have to "regularly consult [...] to make a return on one's portfolio," such as earnings reports, company press releases, or social media. We deliberately chose very strong formulations for the questions, emphasizing that the described behaviors are of essential nature

for successful stock market participation. We therefore interpret our measures as a conservative estimate of the true prevalence of the *active investing* mental model.

Results Column 1 of Table A.6 shows that 76.9% of non-owners agree with the *active investing* mental model compared to only 55.1% of respondents who participate in the stock market, a difference of 39.6%. Non-investors are also 68.8% more likely to believe that one has to monitor one's portfolio at least once a day (Column 2), 53.2% more likely to believe that quick reactions to daily changes in market conditions are necessary (Column 3), and 41.8% more likely to believe that selecting a few stocks that are likely to perform well is the key to successful stock market participation (Column 4). Moreover, non-investors select 36.8% of a standard deviation more public information sources as being required reading when owning stocks (Column 5). As a result, non-investors expect to spend 41.1% more time per week managing their portfolio compared to respondents who own stocks (Column 6). Panel A of Table A.7 shows that these results are not driven by differences in financial literacy, and are robust to controlling for a large set of background characteristics, including education, income, and financial assets. Panel B of Table A.7 further shows that our main survey measure of the active investing model is strongly correlated with its subfacets, supporting the construct validity of the measure.

Taken together, these findings confirm that investors and non-investors have systematically different mental models of the stock market regarding the time and effort cost of investing. This confirms our key qualitative hypothesis from the analysis of our AI-led interviews, providing a proof of concept that AI-conducted interviews can generate hypotheses about economic behavior that can be empirically tested out-of-sample with quantitative methods.

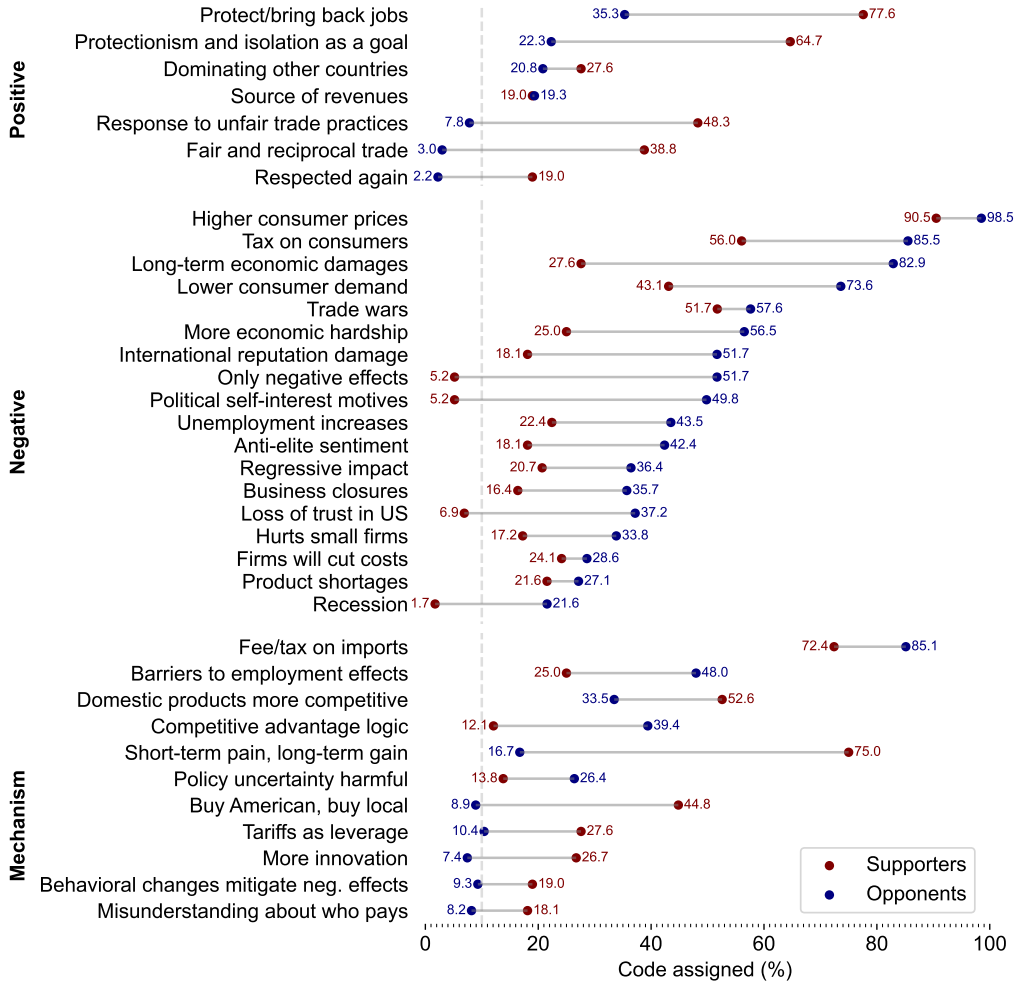
4.3 Main qualitative findings from the tariff interviews

We now turn to our second application on economic policy views. An advantage of this setting is that we can compare reasoning between opponents and supporters of tariffs to see whether the interviews uncover important differences in underlying reasoning.

In contrast to accounts questioning the ability of the American public to reason effectively about public policies (Bartels, 2005), our AI-led interviews reveal that both supporters and opponents of US import tariffs hold rather nuanced views about their effects (see Figure 5 for code frequencies).[19] Rather than being ignorant about the other side's arguments, most respondents acknowledge the short-term negative effects of tariffs—such as rising prices—but differ in how they believe the economy will respond in the long term. What emerges are not simplistic opinions, but competing mental models that reflect divergent beliefs about how firms, consumers, and trade partners will respond to import tariffs.

[19] A comprehensive discussion of the findings from the tariff interviews is beyond the scope of this paper. Instead of striving for narrative completeness in our discussion, we focus on more novel themes and factors that will be important for our validation and benchmarking exercises.

Figure 5: Frequency of discovered codes among supporters and opponents of import tariffs



Note: This figure presents the share of respondents that can be assigned a specific code based on the open-ended text responses they provided (in %). We use data from the tariff interview study (see Table A.2 for an overview of data collections). Codes assigned by AI coding. The vertical lines indicate the 10% code frequency thresholds. Support for US import tariffs elicited during the screener survey prior to the interview.

4.3.1 Opponents

When asked about their views on tariffs, opponents strongly focus on the many perceived short-term costs and long-term macroeconomic risks. They understand tariffs as being a "tax on the American consumer," anticipating that firms will pass on tariff-induced costs to consumers:

> *The reason or reasons that I oppose the new 10% tariffs on imports is that the price increases are passed on to us, the consumer. In this current market, where prices keep increasing, people don't have extra money, and adding tariffs is going to make purchasing everything more expensive, making it more difficult for people to survive.*

For many opponents, these concerns about inflation are compounded by worries about the regressive, distributional effects of tariffs. As one participant notes, tariffs "will affect every class of people, but it will hit the middle class, lower class the hardest." On the supply side, opponents worry about rising unemployment as firms try to cut costs and, as one participant points out,

tariffs will "likely cause many small and medium-size businesses to go out of business because of reduced sales, reduced profits, and uncertainty." As a result, opponents unanimously expect tariffs to cause severe long-term economic damages, describing a possible "domino effect" where higher prices reduce consumer demand, force businesses closures, cause product shortages, and ultimately hurt growth. There is also considerable concern about deteriorating international relations, with opponents expecting other countries to "retaliate, plain and simple. They will not take the threat [of tariffs] lightly and they should not. It will isolate the US." Taken together, the prevailing view among opponents is that tariffs are a self-inflicted wound motivated by political self-interest rather than public welfare.

When asked about potential benefits of tariffs, the most common response is that tariffs "only have negative effects." Yet, even the most skeptical opponents understand what tariff supporters see as the goals of the policy: protecting jobs and reviving manufacturing. The disagreement lies not in the goals themselves, but in whether they are perceived to be achievable. Opponents are quick to point to many structural barriers that, in their view, make job creation unlikely. These include the time required to rebuild capacity ("this is a long, long process"), the absence of industrial infrastructure ("the factories aren't built"), and broader labor market barriers. Specifically, factory work is seen as being unattractive and underpaid—"no one wants to work in factories and there is no way those jobs would pay a living wage"—and the US workforce is viewed as lacking both the training and the willingness to return to manual manufacturing jobs. Even if production came back, respondents expect limited employment gains due to automation. Ultimately, what many opponents invoke are competitive advantage arguments to explain that reshoring jobs would simply not make any sense economically:

> *The workforce in the US is largely, I mean, it's disproportionately white collar to blue collar compared to other countries. So that's why we use our competitive advantage to only manufacture high ticket items or to sell services rather than products. And other countries like China or Vietnam might have a lot of factories where people are willing to go in and work manual labor jobs, or, you know, jobs that create a product rather than just a service. So I see the need for us to take advantage of, you know, our strengths and other countries to take advantage of their own strengths and for us to mutually benefit off of each other.*

4.3.2 Supporters

Tariff supporters, for their part, begin from the same shared understanding of short-term costs—that prices will go up, that there may be short-term disruptions—but are much more optimistic about their potential long-term impacts. While the majority of opponents think tariffs have only negative effects, the large majority of tariff supporters have a "short-term pain, long-term gain" mental model of tariffs, where they expect long-term improvements in economic growth, fairness, and national strength.

Many participants describe tariffs as a means of restoring domestic self-sufficiency and reducing reliance on foreign imports. "We should be able to manufacture medical supplies,

weapons, steel, etc. here in the US," one participant argued, linking tariffs to both economic and national security. Others describe tariffs as tools to "push us to source our own goods from within the US and not other countries" or as part of a broader goal of "making everything in the U.S. and isolating the U.S. from the other parts of the world." In this sense, tariffs are not only protective but a catalyst for restoring American jobs and independence by incentivizing the purchase of goods made in the US.

Closely linked to these policy goals are recurring themes of fairness and reciprocity in global trade. Many supporters feel that the US has long been treated unfairly, pointing to other countries' high tariffs or exploitative labor practices as sources of competitive disadvantage. "For many years other countries charged tariffs on items imported from America, but we did not charge any tariffs to other countries," one participant notes, concluding that "tariffs will assist our country in establishing a level playing field." Another put it more bluntly: "If other countries are charging our country tariffs, then it only makes sense we do the same." These fairness concerns often carry a moral undertone: imposing tariffs is not just economically sensible but "the right thing to do," a way to ensure that "no country is better off than the other by trade."

While the objectives of job creation and fairness in international trade relations are commonly mentioned, what the interviews uniquely reveal are the economic *mechanisms* that supporters invoke to explain why tariffs will work as intended.

First, many supporters believe that tariffs will *increase* innovation. They argue that removing access to cheaper imports will create incentives for US firms to innovate: "Necessity is the mother of invention," one participant wrote: "Because tariffs will reduce the amount of foreign goods flooding US markets, American manufacturers will need to find ways to scale up their production and cut costs while preserving quality." Others highlighted the need to develop more efficient production methods: "It will force companies to produce things locally... to build more advanced factories... which would involve a lot of innovation and change."

Second, participants frequently expressed the view that consumer behavior will mitigate the negative effects of price increases. "Prices of American-made goods would be lower than imported goods," one respondent explained, "and knowing that their purchase would build our American economy would bring satisfaction as well." Several described switching to domestic alternatives, avoiding non-essentials, or seeking higher-quality American products as natural consumer responses. This substitution channel is seen as buffering the cost of tariffs, reducing the real impact on household budgets through endogenous spending responses.

Finally, many supporters see tariffs not as permanent policies but as tactical instruments—a way to exert pressure on other countries and extract better trade deals. Several described tariffs as a "bargaining chip," a "game of chicken," or a means to "push [other countries] into better deals." As one participant put it: "Initially, I think it would cause a lot of chaos and price spikes, but ultimately other countries would come to the negotiating table because they simply can't

maintain a 30% tariff." While both supporters and opponents mention the risk of trade wars, their interpretation differs: for supporters, economic leverage and market size will allow the US to prevail in any retaliatory cycle.

4.3.3 Key findings

Taken together, these interviews show that disagreement over tariffs is less about ignorance and partisan cheerleading and more about disagreement about economic mechanisms. Supporters and opponents often share the same views on short-term economic costs, but differ widely in how they believe economic actors will respond to shape the long-term effects of the tariffs. Opponents focus on structural barriers—missing infrastructure, labor shortages, or global competition and trade wars—that they see as limiting the gains from tariffs. Supporters, by contrast, have a "short-term costs, long-term benefits" view in which they believe that firms will innovate, consumers will adapt, and the US can use its leverage to secure better trade terms. These competing mental models shape not only what respondents expect to happen, but also how they judge whether the policy is worth it.

5 Benchmarking AI-led interviews against survey methods

Having established that AI-led interviews lead to rich and novel insights across relevant economic settings, we now turn to a fundamental methodological question of how AI-led interviews compare to qualitative data collected from surveys with standard open-ended questions. Does the property that qualitative interviews measure "more" than standard survey methods carry over to AI-led interviews? By benchmarking the set of discovered codes in AI-led interviews against those discovered by standard survey methods, we will show that AI-led interviews provide a "double dividend": AI-led interviews discover (i) a broader set of codes in the cross-section and (ii) reveal a broader range of concerns within participants, whereas open-ended survey questions tend to discover a narrower set of codes that are top-of-mind. We also show that AI-led interviews boost text production compared to survey questions.

5.1 Design

Survey We conducted the stock market benchmark survey ($N = 384$, Nov 2024) and the tariff benchmark survey ($N = 249$, July 2025) with US Americans recruited from Prolific using the same screening criteria that we used in the corresponding interview studies (see Section 2.2 for the criteria and Table A.2 for study details). Both benchmark surveys use instructions identical to the interview surveys, except that we remove instructions related to the interview. Instead of the interview, the surveys include a series of six predetermined open-ended questions, presented on separate survey pages. The first open-ended question in the benchmark surveys mirrors the opening question of the respective interview, such as "What are the main reasons you support or oppose the new 10 percent tariff on imports?" for the tariff benchmark. The additional

open-ended questions in the benchmark surveys are designed as "mirrors" to the topics *explicitly* mentioned in the topic guides for the interviews. For example, the second topic for the stock market interview instructed the AI interviewer to "delve into the perceived barriers or challenges preventing them from participating in the stock market." Our corresponding mirror question in the stock market benchmark survey thus becomes: "What barriers or challenges prevent you from participating in the stock market?" In addition, we include all predetermined closing questions from the interviews. Section F.2.3 and F.1.3 present the wording of the mirror questions.

Benchmark 1: Single open-ended question Our first benchmark represents the *status quo* approach of how economists collect open-ended data from study participants (Haaland et al., forthcoming): We compare the codes discovered in the first open-ended survey question of the benchmark survey with the codes discovered when using the full AI-led interview. This allows us to answer the natural question whether researchers would overlook relevant factors that might affect their conclusions when restricting themselves to a stand-alone open-ended question.

Benchmark 2: Multiple open-ended questions Our second benchmark consists of the codes discovered after pooling the open-ended responses at the participant level. The mirror questions allow us to account for the role of the topic guide and the effect of simply asking more open-ended questions. A key remaining difference to an interview is that the benchmark surveys do not include *adaptive* follow-up questions.

5.2 Results

Aggregate results We first study how the frequency and number of *unique* codes that a researcher could "discover" differ between AI-led interviews and the benchmarks of open-ended survey questions. For this analysis, we call a code *discoverable* if at least 10% of participants would be assigned the code by AI coding in an AI-led interview. We then declare a code as *discovered* by a method if it would be assigned to at least 10% of participants.[20] We then examine discovery in the cross-section (extensive margin) and at the individual level (intensive margin).

Table 2 presents the results separately for the stock market (Panel A) and tariff interviews (Panel B). Focusing first on discovery in the cross-section, the single open-ended question discovers only 16–21% of the 29-37 codes discovered with the AI-led interview. Asking more open-ended questions raises the discovery rate to 35–54%, but still half of the discoverable codes remain hidden compared to the AI-led interview. Moving to the participant level, we see a similar picture. With a single open-ended question, one would assign about two codes per participant on average. Asking more open-ended questions increases the number of codes per participant, but it remains below half the level of 9 to 13 codes that an AI-led interview would elicit. This suggests that one would strongly underestimate the richness of how people reason about the

[20] We use this threshold for two reasons. First, AI coding is about 6 pp more liberal on average in assigning codes than human annotators, as discussed in Section 2.3.2. Second, potential codes present in less than 4% of the data are unlikely to be noticed or included in the coding manual of a researcher.

Table 2: AI-led interviews discover "more" than existing survey methods

		Survey benchmark	
	(1) AI-led interview	(2) Multiple open-ended questions	(3) Single open-ended question
A. Economic behavior *(stock market)*			
Discovered codes	29	10	6
Average number of codes per participant	9.1	3.9	2.1
Average frequency of codes	31.3%	13.4%	7.3%
Average frequency of non-top-of-mind codes	25.0%	6.6%	2.7%
Discovery rate	100.0%	34.5%	20.7%
B. Economic policy views *(tariffs)*			
Discovered codes	37	20	6
Average number of codes per participant	13.1	6.7	2.0
Average frequency of codes	35.5%	18.2%	5.4%
Average frequency of non-top-of-mind codes	32.3%	14.0%	2.9%
Discovery rate	100.0%	54.1%	16.2%

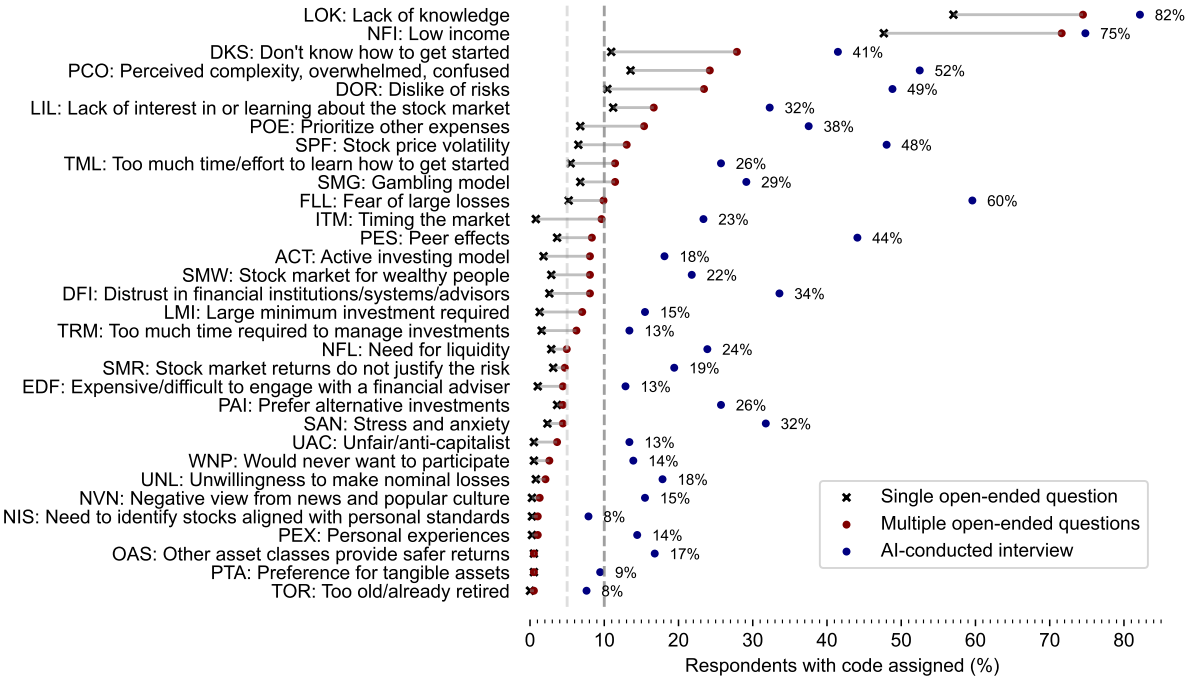
Note: This table presents summary statistics for discovered and assigned codes across data collection methods and applications. The table combines data from the stock market (Panel A) and tariff interviews (Panel B), and the corresponding benchmark surveys with pre-determined open-ended questions (see Table A.2 for an overview of studies). Column 1 presents statistics for AI-led qualitative interviews. Column 2 presents statistics for a benchmark of asking multiple pre-determined open-ended survey questions that cover all topics included in the topic guide of the AI-led interview. Column 3 presents statistics for the benchmark of asking a single open-ended survey question. For comparability, statistics are based on AI coding. We exclude infrequent codes that are assigned to less than 10% of participants in an AI-led qualitative interview. "Discovered codes" is the number of codes that can be assigned to at least 10% of participants. "Average number of codes per participants" is the average number of unique codes assigned to a participant. "Average frequency of codes" is derived by computing for each code the share of participants that can be assigned this code, and then averaging this number across all codes. "Average frequency of non-top-of-mind codes" is analogously derived by excluding codes that can be assigned to more than 10% of participants when asking only a single open-ended question. "Discovery rate" is the share of discovered codes relative to an AI-led interview.

interview topic with open-ended survey questions without adaptive probing. On average, code frequencies are about 17 percentage points higher in AI-led interviews compared to multiple open-ended survey questions. Notably, the gains from additional open-ended questions skew towards the "top-of-mind codes" that are already prevalent in a stand-alone open-ended question. If we exclude these codes, the average code frequency is low at 6.6% to 14%, even when asking multiple open-ended survey questions.

Taken together, this suggests that AI-led interviews reveal substantially more unique codes, themes, and factors than responses to standard open-ended survey questions. We next turn to the level of an individual code to characterize the *nature* of codes that remain hidden—unless we conduct an AI-led interview. We present results separately by application.

Code-level results: Stock market Turning first to the stock market interviews, Figure 6 plots code-level frequencies derived from AI coding separately for AI-led interviews and our two benchmarks. Starting with our benchmarks, we find that a single open-ended question on why people are not participating in the stock market would revolve around knowledge barriers (*LOK*)

Figure 6: Stock market non-participation: AI-led interviews compared to survey benchmarks



Note: This figure presents the share of respondents that can be assigned a specific code based on the open-ended text responses they provided (in %). We combine data from the interview sample of our stock market interview study and the stock market benchmark survey (see Table A.2 for an overview of data collections). All participants answered open-ended questions about their reasons for not participating in the stock market. "AI-conducted interviews" presents code frequencies based on the full AI-led interview transcripts from our main experiment (i.e., whether a given code was assigned at least once based on the full interview). "Multiple open-ended questions" presents code frequencies based on the responses to six open-ended survey questions in our benchmark survey. These open-ended questions correspond to the topic guide of the main experiment and thus cover the same topics as our main interview, except for any follow-up questions that an interviewer might have asked. "Single open-ended question" presents code frequencies based *only* on the response to the first open-ended question in our benchmark survey. All codes used for this exercise are derived from AI coding for comparability. The vertical lines indicate the 5% and 10% code frequency thresholds.

and income constraints (*NFI*). When asking multiple open-ended survey questions, complexity perceptions (*PCO*) and general risk aversion (*DOR*) would additionally surface as major themes.

Against these benchmarks, the first broad conclusion is that researchers using an AI-led interviews would additionally discover conceptually distinct *classes* of codes. First, and most notably, AI-led interviews would reveal codes tied to participants' *mental models* (*ACT*, *SMG*, *ITM*) that are very difficult to detect with standard open-ended survey questions. This is mirrored in the participation costs codes related to the "active investing model" (*TML*, *TRM*) that only surface during interviews. In our context, conclusions about the relevance of (misperceived) participation costs would strongly depend on whether a researcher is conducting interviews or relying on open-ended survey questions. Another conceptually distinct class of codes that tends to only show up during an AI-led interview are social influences, such as peer effects (*PES*), the role of personal experiences (*PEX*), or emotions (*SAN*).

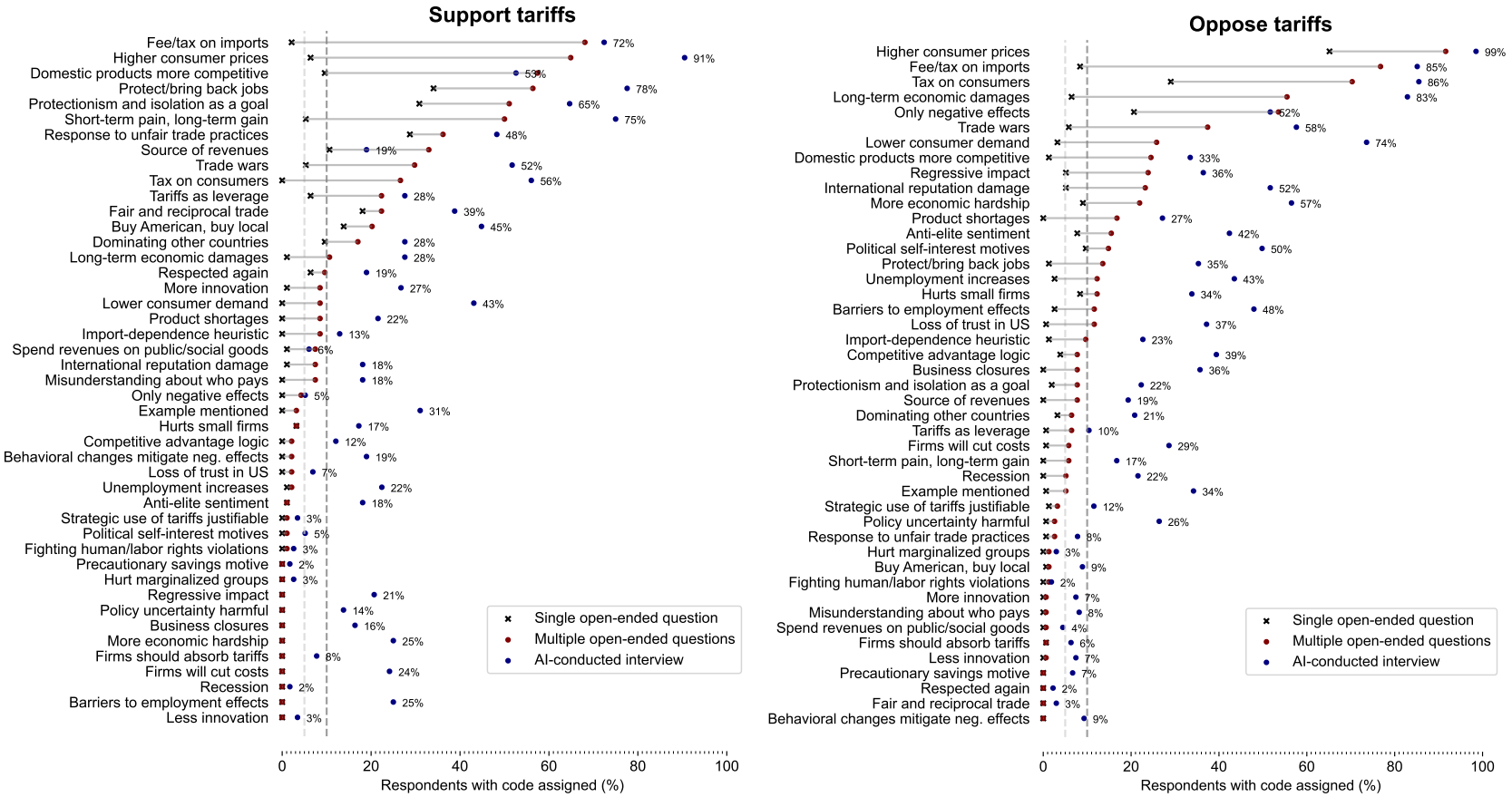
The second finding is that AI-led interviews meaningfully change the *order* of code frequencies. For example, the third most common code in AI-led interviews—"fear of large losses" (60%)—is only assigned to 5% of participants based on a single open-ended survey questions. Similar patterns hold for social influences (*PEX*) or codes tied to emotions (*SAN*). This suggests that interviews are not mechanically increasing code frequency, but can reveal highly prevalent factors that are simply not on top of people's minds when responding to open-ended survey questions; and that merely asking "more" open-ended survey questions may not be an effective substitute for AI-led interviews.

Code-level results: Tariffs We replicate the above finding in the context of our import tariff interviews: AI-led interviews can reveal people's underlying *mental models* and economic *mechanisms* that are largely absent from open-ended survey questions without adaptive probing.

Starting with tariff opponents, Figure 7 shows that responses to a single open-ended question would reveal the "higher consumer prices," "tax on consumers" and "only negative effects" codes. While asking additional open-ended questions elicits some additional concerns, only the AI-led interviews reveals key mechanisms that explain *why* opponents are skeptical about potential benefits of tariffs (e.g., "barriers to employment effects," "competitive advantage logic"). Among tariff supporters, we similarly find that key mechanism-based arguments are largely absent from survey responses to open-ended questions. For example, without an AI-led interview, researchers might struggle to detect reasoning related to innovation ("more innovation") or consumer substitution ("behavioral changes mitigate neg. effects") that explain *why* supporters are not as concerned about the short-term costs of tariffs. Similarly, using data from open-ended survey questions alone would suggest that supporters almost exclusively see positive, while opponents almost exclusively see negative aspects of tariffs. Yet, when conducting a conversation, both sides express awareness of the key arguments posited by the other side; in other words, the opposing views on tariffs revolve much more around *disagreement* about which mechanisms will dominate in the long term rather than ignorance about the other side's views. Taken together, these results underscore that AI-led interviews not only elicit a broader set of themes but are essential for uncovering how individuals reason about economic policies, and why they come to different conclusions.

Dynamics of discovery The above finding that AI-conducted interviews reveal novel, relevant factors that would be largely unobserved when using open-ended survey questions raises a natural question about timing: When are codes discovered? In Appendix Section C.1, we analyze how the discovery of different themes unfolds over the course of an interview. We find that more "mechanical" explanations, such as those focusing on economic constraints (e.g., low income), are revealed within the first few questions, while higher-order objects—such as mental models and broader economic reasoning—appear more gradually over time.

Figure 7: Tariffs: AI-led interviews compared to survey benchmarks



Note: This figure presents the share of respondents that can be assigned a specific code based on the open-ended text responses they provided (in %). We combine data from the tariff interview study and the corresponding benchmark survey (see Table A.2 for an overview of data collections). "AI-conducted interviews" presents code frequencies based on the full AI-led interview transcripts from our main experiment (i.e., whether a given code was assigned at least once based on the full interview). "Multiple open-ended questions" presents code frequencies based on the responses to six open-ended survey questions in our benchmark survey. These open-ended questions correspond to the topic guide of the main experiment and thus cover the same topics as our main interview, except for any follow-up questions that an interviewer might have asked. "Single open-ended question" presents code frequencies based *only* on the response to the first open-ended question in our benchmark survey. All codes used for this exercise are derived from AI coding for comparability. The vertical lines indicate the 5% and 10% code frequency thresholds.

5.3 Text-production and mode of interview

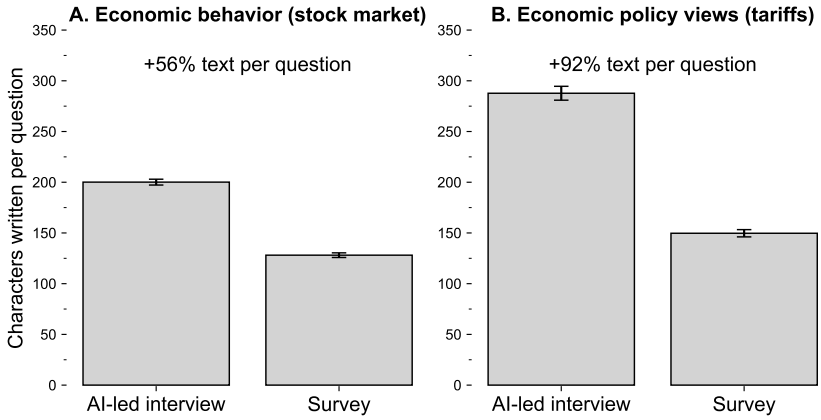
Text-production A common issue with open-ended responses in surveys is that they are often short and thus lack the necessary context for text analysis. It is not uncommon that a sizable fraction of survey responses cannot be classified due to the brevity or ambiguity of the responses, thus rendering the data unusable for subsequent analysis. AI-led interviews might be more engaging, which could boost participants' willingness to write more text, even when holding the *number of questions* constant. To test this, we combine data from the interview and the benchmark surveys. From the AI-led interviews, we retain the responses to the first question of each topic of the interview guide, i.e., the questions for which we have a "mirror" question in our benchmark surveys. We then calculate the grand mean of the number of characters written across participants and questions, separately by method and application.

Figure 8 presents the results. The stock market interviews increased text production on a per-question basis by 56% (+72 characters) against a baseline of 128 characters in the benchmark survey (Panel A). For the interviews about economic policy views—which were conducted with a more advanced LLM (`GPT-4o`)—text production is 92% higher (+138 characters) against a baseline of 156 characters in the benchmark survey. Table A.8 replicates these results on a question-by-question basis, including the very first question of the interview.[21] More surprisingly, we also find that text production is stable over the course of an entire AI-led interview, as shown in Figure 2. These findings suggest that the conversational nature of AI-led interviews can help overcome a key limitation of open-ended survey questions.

Interview mode: Voice vs typed input During our import tariff interviews, participants could respond either by typing or by recording a voice message. While this choice is endogenous, it offers interesting correlational evidence on the relevance of the response mode. 39% used the voice message functionality at least once, and Table A.9 presents descriptive statistics by input mode. The time participants take completing the interview is virtually identical with 27 minutes for both input modes on average (Column 1), and we detect no difference in interview attrition rates across input modes ($p = 0.165$). As expected, people can talk faster than they can write: text production is 112% higher with voice input compared to the 3,430 characters participants typed on average (Column 2). Perhaps surprisingly, but mirroring the findings from randomized trials (Namey et al., 2020), this substantial difference in text does not necessarily translate into richer responses: the number of unique codes assigned from our coding manual is only 8% higher ($p = 0.009$) with voice input (14.3 codes) compared to typing (13.1). Moreover, Figure B.5 shows that the frequency of individual codes is largely similar across modes. A notable exception is that voice input users are more likely to mention specific examples, corroborating the findings from Galasso et al. (2024). While response mode will surely matter in some contexts and for

[21] Note that participants could not copy-paste text from AI tools such as ChatGPT into the chat interface due to programmatic restrictions added for the tariff interviews, which rules out that these results are driven by AI use. For the stock market interviews, self-reported weekly usage of AI tools is uncorrelated with text production (p = 0.815).

Figure 8: AI-led interviews boost text production compared to open-ended survey questions



Note: This figure shows the mean number of characters written in response to a question in an AI-led interview and in response to an open-ended survey question of equivalent nature. For this comparison, we combine data from the interview studies and the benchmark surveys (see Table A.2 for an overview of data collections). From interviews, we keep the opening question and the first question from each interview topic. These questions were designed to have mirror questions in the benchmark survey. We exclude the predetermined closing questions. See Section 5.1 for details. Standard error of the mean shown as error bars.

specific participant population, it is reassuring that the practical differences might be muted in economic applications where participants can actively choose the mode they prefer.

6 Internal and predictive validity of AI-led interviews

Our final exercise addresses a natural question many economists used to studying revealed preference data might have: How meaningful are the responses obtained from AI-led interviews? Can we trust that the conversations give meaningful insights to predict and understand economic preferences and behaviors? Similar concerns were once raised about survey methods, with skepticism about whether self-reports could yield valid insights into preferences or behavior. We anticipate that this debate will be revisited in the context of AI-led interviews, where a potential concern is that interviewees just try to fill the conversation space created by the AI interviewer. While sociologists typically see the primary value of interviews as in generating novel ideas, economists frequently seek an additional output: measurements that can be used for quantitative analysis. This motivates the current section. We ask whether the codes derived from AI-led interviews correlate with conceptually related survey measures (internal validity), and whether they can predict incentivized behavior in a long-term follow-up (predictive validity).

6.1 Internal validity: Correlations with survey measurements

If participants provide meaningful and informative responses, we would expect a correlation between codes assigned to interview transcripts based on our coding manual and direct mea-

surements of conceptually related constructs based on standard survey questions.[22] To test this, both the stock market and the tariff interview studies included a battery of survey questions after the interview module. In the stock market study, we presented participants with a multiple choice list of common reasons for stock market non-participation in randomized order and asked participants to select all that apply. In the tariff study, we elicited expectations about the effects of imposing import tariffs using 5-point Likert scale questions (e.g., macroeconomic or distributive effects), which we standardize. For each study, we match survey questions with a theoretically matching code from our coding manual. For example, in the tariff case, we match the survey question eliciting beliefs about whether low-income households will gain or lose with the *regressive impact* code from our coding manual. We can match 14 (12) survey question from the stock market (tariff) studies with a code from our coding manuals. While we tried to include a comprehensive survey battery of structured Likert questions, the incomplete matching of coding illustrates that it is difficult to predict ex-ante all the relevant factors that will surface in the interviews.

Figure 9 presents the results from separately regressing the survey measures on a dummy for the presence of their matched code. While there is variation in the strength of association, correlations are sizable and in the expected direction. Panel A shows that participants are, on average, 45 percentage points more likely to select a reason for stock market non-participation from the multiple choice question if they have been assigned the matched code. Panel B shows that participants that are assigned a code score, on average, 0.52 standard deviations higher on the matched survey question.

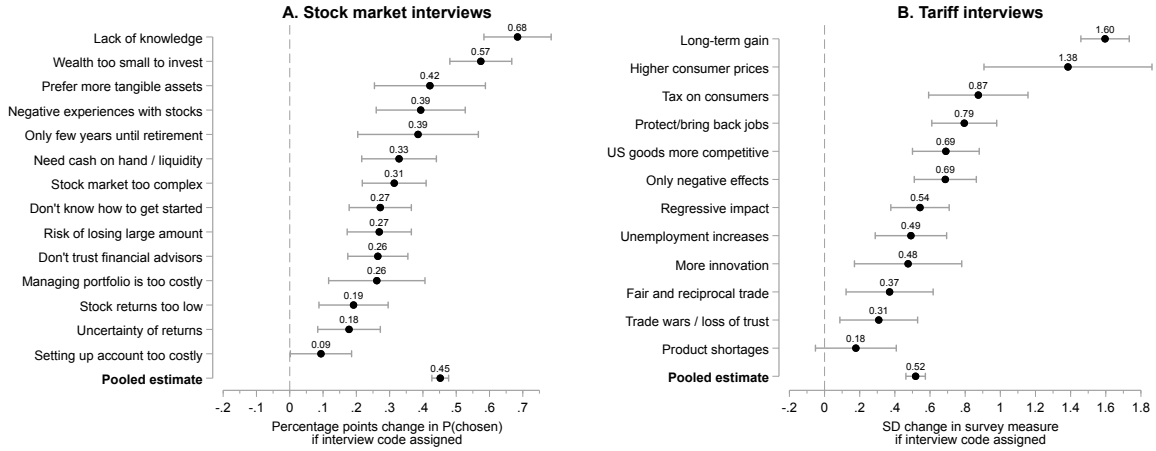
For the stock market interviews, we can additionally test whether interview codes are correlated with validated survey measures of *economic traits*, such as preferences, beliefs, and constraints. Table A.10 presents the results from this exercise. For example, Column 1 shows that participants who are assigned the *dislike of risk* code are 0.55 standard deviations more risk-averse using a validated risk preference measure (Falk et al., 2018). We find similarly pronounced associations for other economic factors, such as financial literacy (Column 2), subjective beliefs (Column 3), measures of constraints (Column 4–5), or asset class preferences (Column 6). Taken together, these findings demonstrate the internal and convergent validity of measurements derived from AI-led interviews.

6.2 Predictive validity: Behavior in a long-term follow-up

A complementary, and perhaps more demanding, question specific to the context of interviews with the goal of understanding economic behavior is whether participants' implicitly or explicitly *stated explanations* during an AI-led interview are behaviorally predictive. To address this, we

[22] Our goal is to show that the measurements are correlated. We take no stance on which measurement is methodologically preferred or a more valid "ground truth." For example, if researchers know precisely what they would like to measure, a tailored survey question might be better suited. If measurement should avoid priming a particular mechanism or if discovering unexpected mechanisms is key, an interview might be preferred.

Figure 9: Internal validity: Interview codes correlate with direct survey measurements



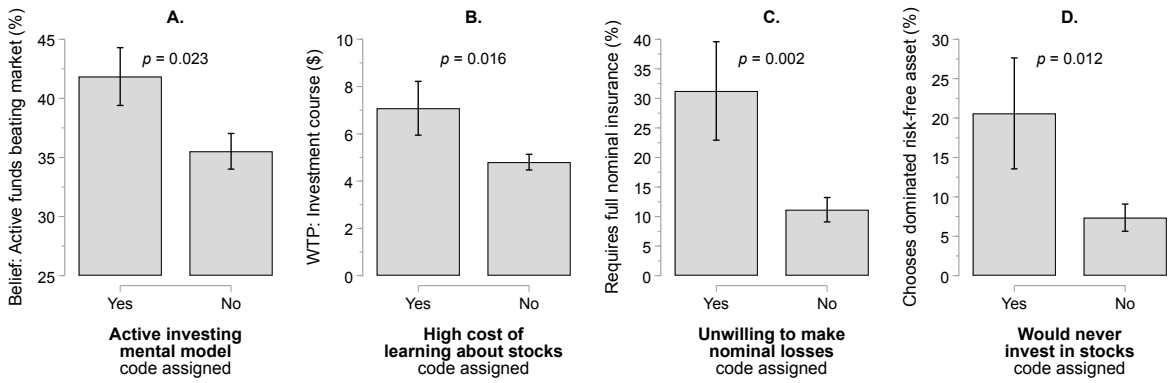
Note: This figure combines bivariate regression estimates from the stock market (Panel A) and tariff (Panel B) interview studies. We match post-interview multiple choice response options (Panel A) or 5-point Likert scale questions (Panel B) with a conceptually related code from our coding manual. In Panel A, we regress an indicator for having selected a specific multiple choice option (0/1) on a dummy for the presence of the code (0/1). In Panel B, we regress the standardized Likert scale response on a dummy for the presence of the code (0/1). In addition, each panel represents a pooled estimate when reshaping the data to the interview-by-survey measure level. Labels in Panel A correspond to the label of the multiple-choice option, while labels in Panel B correspond to the code matched to the Likert scale. We use codes assigned by AI coding. 95% CIs derived from robust standard errors are shown.

conduct a long-term follow-up with our stock market interview sample eight months later that includes incentivized behavioral outcomes.

Design We invite stock market interview participants from Aug/Sep 2023 to a long-term follow-up survey in May 2024, and manage to recontact 70% of the original sample. Table A.3 provides summary statistics and Appendix Section F.1.4 contains the experimental instructions. A key design objective is generating testable behavioral predictions for a subset of codes from our coding manual. To focus on non-participation motives that only show up later during interviews, we selected four codes that are almost never assigned to initial responses: *active investing model* (*ACT*, 0.5% discovery rate in the first response, see Figure 4), *high learning cost* (*TML*, 1.8%), *unwilling to make nominal losses* (*UNL*, 0.5%), *would never participate* (*WNP*, 0%).[23] The survey includes three incentivized outcomes tailored to these interview codes. First, we elicit respondents' incentivized belief about the share of actively managed funds that yielded higher after-fee returns than passively managed funds in 2023 on a scale from 0 to 100%. Respondents could earn a bonus of $2 if they guess within 2 percentage points of the actual value. This allows us to test whether respondents who revealed an "active investing" mental model during interviews believe that active portfolio management outperforms passive investing. Second, we elicited respondents' probabilistically incentivized willingness to pay (WTP) for three-month access to an online "four-part series aimed at helping beginners understand the basics of stock

[23]When regressing a dummy for participating in the follow-up survey on a set of dummies for whether these codes are assigned, we find no evidence of differential attrition. The p-value of a joint F-test is 0.416.

Figure 10: Interview responses predict incentivized behavior eight months later



Note: Data from the long-term stock market interview follow-up (see Table A.2 for a study overview). Each panel separately shows the mean and standard error of an incentivized decision for participants that were either assigned a certain interview code from our coding manual, or not. Panel A: Belief about the share of actively managed funds that outperform passively managed funds. Panel B: WTP for a 3-month online investment course. Panel C: Share of respondents who only switch from a certain payment of $1,000 in 12 months to investing $1,000 in the S&P 500 for 12 months if offered full nominal insurance. Panel D: Share of respondents who are unwilling to invest in the S&P 500 even if offered full nominal insurance. p-values for t-tests for equality of means are shown.

market investments and learn how to successfully navigate the market" from MasterClass, a platform providing high-quality learning content. We elicited the WTP using a multiple price list ranging from $0 to $20 in increments of $2. It allows us to test whether respondents who worry about high learning costs, both in terms of time and effort, have a higher WTP for an investment course designed to lower these barriers. Third, we elicited respondents' demand for loss protection in the stock market using a multiple price list. Respondents repeatedly choose between two options. Option A is a certain payment of $1,000 in twelve months. Option B grants the value of $1,000 invested in the S&P 500 in twelve months. Additionally, Option B caps the maximum loss from this investment ("loss protection"). Across the price list, we vary the insurance against nominal losses from 0% to 100%, i.e., we are capping losses at the indicated percent value. We implement a randomly selected choice for one out of every 200 respondents. This outcome provides two predictions. Respondents who reveal an "unwillingness to make nominal losses" (e.g., "I would need a *guarantee* on my investment") should *only* switch away from the risk-free Option A if offered *full* insurance against nominal losses. In addition, people who emphasize that they would under "no circumstances" invest in the stock market should choose the risk-free option even if offered 100% insurance.

Results Panel A of Figure 10 shows that respondents with an *active investing* mental model report a 6.4 percentage points higher belief about the share of active funds that outperform the market compared to those who are not assigned the code ($p = 0.023$). Panel B shows that the average WTP for an investment course is 48% higher than the WTP among respondents who do not mention high learning costs as a barrier to investing in stocks ($p = 0.016$). Panel C shows

that the probability of switching from the certain payment to an investment in the stock market *only if* offered full insurance against nominal losses increases by a factor of 2.8 if respondents reveal an unwillingness to invest in assets that include the possibility of "ending up with less than one started with" ($p = 0.002$). Finally, we observe that respondents who were assigned the *would never invest* code are more likely to choose the dominated certain payment over an investment in the S&P 500 with a guaranteed minimum return of 0% ($p = 0.012$).

Taken together, our results demonstrate that AI-led interviews capture meaningful and behaviorally predictive factors. Interview codes not only align closely with direct survey measures but also predict incentivized choices eight months later, mitigating concerns about "cheap talk" and highlighting their value for understanding economic behavior.

7 Conclusion

This paper proposes the first scalable method for interviewing human subjects by delegating the task of interviewing to an AI interviewer. Our AI interviewer application allows researchers to run unlimited interviews in parallel at effectively zero marginal cost beyond participation incentives, making AI-led qualitative interviews equally cost-effective as traditional surveys. We provide a comprehensive, multidimensional evaluation of AI-led interviews, demonstrating across two complementary settings that our novel method allows survey researchers to gather richer qualitative data without increasing costs.

First, we show that the AI interviewer follows methodological interviewing guidelines and displays contextual awareness when crafting interview questions. Second, the AI interviewer strongly engages participants in a natural conversation that leads to high participant satisfaction and higher text production compared to open-ended survey questions. Third, we show that qualitative data generated from AI-led interviews allow hypothesis generation, the hallmark feature of qualitative interviews. Fourth, AI-led interviews outperform existing survey methods in economics for collecting qualitative data. Notably, we show that AI-led interviews reveal insights that are often absent from open-ended survey questions, such as mental models and social influences. Finally, we show that AI-led interviews elicit behaviorally relevant factors, with interview responses predicting incentivized choices more than half a year after the interviews.

The scalability of AI-led interviews will facilitate many novel applications of interviews in economics. For instance, AI interviews can be used to design tailored information provision treatments where the barriers and constraints identified during the interview can be used to design an information treatment for maximum impact. Furthermore, while it is currently standard practice in economics to include some open-ended questions in large-scale surveys, very short AI-conducted interviews—where an open-ended question is followed by one or two probing questions—can increase the quality of open-ended survey data by clarifying potential ambiguities in the initial top-of-mind response.

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Conducting Qualitative Interviews with AI

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September 3, 2025

Section A contains additional tables.

Section B contains additional figures.

Section C presents additional results.

Section D contains additional details about AI-led interviews and a discussion of considerations for conducting AI-led interviews.

Section E contains details about the coding of interview transcripts.

Section F presents the key experimental instructions for our surveys and experiments.

A Additional tables

Table A.1: Interview topic guide for the AI interviewer

	Individual economic behavior: Stock market interviews	**Economic policy views**: US import tariffs
Topic 1 (Opening question, pre-determined)	I am interested in learning more about why you currently do not own any stocks or stock mutual funds. Can you help me understand the main factors or reasons why you are not participating in the stock market? **(6 questions)**	First of all, what are the main reasons you support or oppose the new 10 percent tariff on imports? **(6 questions)**
Topic 2	Delve into the perceived barriers or challenges preventing them from participating in the stock market. **(5 questions)**	Counter-perspective prompt - invite them to consider the side that hasn't been emphasized yet (costs vs. benefits). Example opening question: "So far, we've mostly discussed [dominant theme] effects. Can you think of any potential [opposing theme] effects of tariffs?" **(2 questions)**
Topic 3	Explore a 'what if' scenario where the interviewee invests in the stock market. What would they do? What would it take to thrive? Probing questions should explore the hypothetical scenario. **(3 questions)**	Reasoning about tariffs - invite them to explain how they think tariffs work. Example opening question: "Imagine you were going to explain tariffs to someone unfamiliar with them and how they might affect different groups. How would you explain tariffs to them?" **(4 questions)**
Topic 4	Probe for conditions or changes needed for the interviewee to consider investing in the stock market. **(2 questions)**	What-if scenario - Suppose the United States suddenly raised tariffs to 30 percent on imports from all major trading partners. What do you think would happen, both initially and further down the road? **(2 questions)**
Topic 5	–	Wrap-up - drawing on the full conversation, clarify unresolved ambiguities and deepen promising themes that deserve more exploration. **(3 questions)**
Closing questions (pre-determined)	(1) As we conclude our discussion, are there any perspectives or information you feel we haven't addressed that you'd like to share? (2) Reflecting on our conversation, what would you identify as the main reason you're not participating in the stock market?	Is there anything you would want to know or see that could change your mind about the tariffs?

Note: This table provides an overview of the interview structure and topic guide for our applications (see Section 2.2 for details). The table provides the *exact instructions* from the AI interviewer's prompt on which aspects to focus at each stage of the interview. These instructions are provided to both the topic agent and the probing agent. We indicate the instructed number of questions per topic in bold. The first question of the interview ("opening question") and the closing questions of the interview are identical across participants.

Table A.2: Overview of data collections

Study and population	*Recruitment*	*n*	*Study description*
Stock market interviews			
General population (US), non-stockowners	Prolific, Aug-Sep 2023, screener survey for recruitment	381	Descriptive survey with an integrated 30-minute text-based interview led by an AI interviewer. Post-interview survey questions about the interview experiences. Additional survey modules on factors associated with stock market non-participation, respondent characteristics and household finances.
Long-term interview follow-up survey			
Stock market interview participants	Prolific, May 2024	266	Long-term follow-up with participants from the stock market interview study. Descriptive survey that includes incentivized outcomes to study whether selected interview codes predict choices: (i) demand for insurance against nominal losses, (ii) willingness to pay for an investment course, (iii) incentivized belief about the performance of actively managed funds.
Out-of-sample validation survey			
General population (US)	Prolific, May 2024	1,000	Descriptive survey with equal share of stockowners and non-owners. Survey questions aimed at measuring whether respondents hold the "active investing" mental model that we hypothesized based on the qualitative analysis of the interview transcripts from the stock market interviews.
Selection experiment			
General population (US)	Prolific, Nov 2023	499	Respondents are asked to select all studies that they would like to participate in if we run it: (i) 40-minute survey, (ii) 40-minute interview with a human interviewer, (iii) 40-minute interview with an AI interviewer. These studies are shown on separate pages in randomized order.
Stock market benchmark survey			
General population (US), non-stockowners, recruitment identical to stock market interviews	Prolific, Nov 2024	384	Respondents are asked to answer six pre-determined open-ended survey questions about their reasons for stock market non-participation. The first question is identical to the opening question of our AI-led interviews; the other open-ended questions mimic the topic guide of our interviews.
Tariff interviews			
General population (US)	Prolific, May-Jun 2025, screener survey for recruitment	385	Descriptive survey with an interview with an AI chatbot about US import tariffs from April 2025. Post-interview survey questions about the interview experience. Additional survey questions about tariffs.
Tariff benchmark survey			
General population (US), recruitment identical to tariff interviews	Prolific, July 2025	249	Survey with six pre-determined open-ended survey questions about their views on tariffs, following the topic guide of the tariff interview study.

Table A.3: Summary statistics for the stock market studies

	(1) Interview sample	(2) Long-term follow-up	(3) Benchmark survey	(4) Validation survey	(5) Selection study
Age	39.36	42.24	34.35	37.15	42.63
	(14.03)	(13.87)	(11.48)	(12.59)	(14.97)
Female	0.64	0.62	0.69	0.50	0.49
	(0.48)	(0.49)	(0.46)	(0.50)	(0.50)
College degree	0.53	0.56	0.52	0.58	0.72
	(0.50)	(0.50)	(0.50)	(0.49)	(0.45)
Full-time employment	0.47	0.52	0.37	0.46	0.52
	(0.50)	(0.50)	(0.48)	(0.50)	(0.50)
Caucasian/White	0.77	0.79	0.65		
	(0.42)	(0.41)	(0.48)		
African American/Black	0.12	0.13	0.13		
	(0.33)	(0.33)	(0.34)		
Hispanic	0.11	0.11	0.14		
	(0.31)	(0.31)	(0.35)		
Northeast	0.18	0.17	0.21	0.21	0.26
	(0.38)	(0.37)	(0.41)	(0.41)	(0.44)
Midwest	0.24	0.21	0.21	0.18	0.22
	(0.43)	(0.41)	(0.41)	(0.38)	(0.41)
West	0.17	0.18	0.21	0.25	0.15
	(0.38)	(0.38)	(0.41)	(0.43)	(0.36)
South	0.41	0.44	0.37	0.36	0.38
	(0.49)	(0.50)	(0.48)	(0.48)	(0.48)
Household income (thousands)	71.91	69.29	72.94	76.07	84.70
	(38.76)	(36.91)	(38.42)	(53.94)	(53.01)
Financial assets (thousands)	77.98	82.71	55.34	104.81	
	(138.65)	(145.38)	(114.32)	(162.19)	
Non-mortgage debt (thousands)	27.67	22.73	29.75	22.18	
	(60.06)	(51.22)	(54.31)	(47.37)	
Homeowner	0.51	0.52	0.38	0.46	
	(0.50)	(0.50)	(0.49)	(0.50)	
Home value (thousands)	243.01	239.49	312.84		
	(138.07)	(135.65)	(148.93)		
Mortgage debt (thousands)	124.29	112.82	156.96		
	(91.28)	(78.64)	(128.89)		
Two months of liquid savings	0.60	0.61	0.55	0.68	
	(0.49)	(0.49)	(0.50)	(0.47)	
Observations	379	266	384	1,000	499

Note: This table presents the mean and standard deviation (in brackets) of a range of background variables for the stock market studies (see Table A.2 for an overview of studies), as indicated in the column title. "Home value" and "Mortgage debt" are conditional on non-zero values. Empty cells indicate that the variable was not collected as part of the study.

Table A.4: Summary statistics for the import tariff studies

	(1) Interview sample	(2) Benchmark survey
Age	43.76	45.31
	(13.56)	(13.73)
Female	0.58	0.53
	(0.49)	(0.50)
College degree	0.57	0.55
	(0.50)	(0.50)
Full-time employment	0.45	0.57
	(0.50)	(0.50)
Caucasian/White	0.76	0.72
	(0.43)	(0.45)
African American/Black	0.11	0.22
	(0.32)	(0.42)
Hispanic origin	0.11	0.05
	(0.31)	(0.22)
Northeast	0.19	0.18
	(0.39)	(0.38)
Midwest	0.18	0.22
	(0.38)	(0.42)
West	0.21	0.11
	(0.41)	(0.32)
South	0.42	0.48
	(0.49)	(0.50)
Democrat	0.45	0.41
	(0.50)	(0.49)
Republican	0.30	0.37
	(0.46)	(0.48)
Independent	0.25	0.22
	(0.43)	(0.42)
Supports tariffs	0.30	0.38
	(0.46)	(0.49)
Opposes tariffs	0.70	0.62
	(0.46)	(0.49)
Observations	385	249

Note: This table presents the mean and standard deviation (in brackets) of a range of background variables for the import tariff studies (see Table A.2 for an overview of studies), as indicated in the column title.

Table A.5: No differential selection into AI interviews compared to surveys

	Dependent variable: Demand for study participation (binary)					
	(1)	(2)	(3)	(4)	(5)	(6)
AI interview	0.005	-0.022	0.007	0.003	0.007	-0.010
	(0.034)	(0.060)	(0.035)	(0.049)	(0.049)	(0.075)
AI interview x Willing to write		0.036				
		(0.072)				
AI interview x Stock non-participation			-0.014			
			(0.116)			
AI interview x Above median age				0.004		
				(0.068)		
AI interview x Female					-0.012	
					(0.067)	
AI interview x College						0.009
						(0.084)
N	333	333	333	333	332	332
R^2	0.000	0.003	0.008	0.000	0.001	0.007
Mean of dep. var.	0.895	0.895	0.895	0.895	0.898	0.898

Note: This table shows regression estimates using data from the selection experiment (see Table A.2 for an overview of studies). The dependent variable is a dummy for whether respondents would like to participate in the described study. All regressions restrict to the first study described to respondents, using vignettes that described an AI-led interview or a survey study. "AI interview" is a dummy for whether the study was described as including an AI-led interview, or not. "Willing to write" is a dummy for writing at least 20 words in a screener survey. "Stock non-participation" is a dummy for non-investors. "Above median age" is a dummy for having above median age. "Female is a dummy for female respondents. "College" is a dummy for having completed a college degree. Robust standard errors are shown in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Out-of-sample replication of the active investing mental model in a large-scale survey

	Dependent variable:					
	(1) Active trading mental model	(2) Daily monitoring important	(3) Quick reactions to market important	(4) Actively picking winner stocks important	(5) Public info required to read (z-scored)	(6) Log weekly portfolio management time
Non-owner	0.218***	0.132***	0.173***	0.207***	0.368***	0.411***
	(0.029)	(0.027)	(0.031)	(0.030)	(0.062)	(0.073)
Constant	0.551***	0.192***	0.325***	0.495***	-0.182***	0.575***
	(0.022)	(0.018)	(0.021)	(0.022)	(0.043)	(0.053)
N	1,000	999	999	999	999	999
R^2	0.053	0.023	0.031	0.045	0.034	0.031

Note: Data from the out-of-sample validation survey for the stock market interviews, conducted in May 2024 (see Table A.2 for a study overview). The dependent variables in columns 1–4 are binary variables. "Non-owner" is a dummy for respondents without any exposure to the stock market. "active investing mental model" is a dummy for agreement with the active investing view. "Daily monitoring important" is a dummy for respondents selecting "daily" or "multiple times a day" when asked about how often it is necessary to monitor market conditions and check one's portfolio. "Quick reactions to market important" is a dummy for selecting "the key to stock market success is to react quickly to daily market changes." "Actively picking winner stocks important" is a dummy for selecting "the key to stock market success is to actively pick a few stocks that will increase in value and avoid stocks that will decrease in value." "Public info required to read" is a z-scored number of public information sources that respondents think one has to "regularly consult when participating in the stock market to make a return on one's portfolio." "Log weekly portfolio management time" is the log of the weekly hours that respondents think one would have to spend managing one's portfolio, measured on an interval scale from "less than 1 hour per week" to "more than 15 hours per week." Table A.7 shows that results are robust to controlling for financial literacy and sociodemographic variables. Robust standard errors are shown in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Out-of-sample replication: Robustness to controlling for financial literacy and sociodemographic characteristics

	Dependent variable:					
	(1) Active trading mental model	(2) Daily monitoring important	(3) Quick reactions to market important	(4) Actively picking winner stocks important	(5) Public info required to read (z-scored)	(6) Log weekly portfolio management time
Panel A.						
Non-owner	0.176***	0.120***	0.106***	0.155***	0.334***	0.334***
	(0.032)	(0.031)	(0.033)	(0.034)	(0.069)	(0.082)
Financial literacy (0-3)	-0.113***	-0.072***	-0.124***	-0.089***	-0.070*	-0.225***
	(0.018)	(0.019)	(0.020)	(0.018)	(0.040)	(0.043)
N	999	999	999	999	999	999
R^2	0.098	0.051	0.099	0.089	0.045	0.086
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Panel B.						
Active trading model		0.307***	0.387***	0.301***	0.783***	1.210***
		(0.024)	(0.028)	(0.033)	(0.065)	(0.068)
Financial literacy (0-3)		-0.040**	-0.082***	-0.059***	0.010	-0.093**
		(0.018)	(0.020)	(0.019)	(0.040)	(0.040)
N		999	999	999	999	999
R^2		0.139	0.219	0.148	0.151	0.291
Controls		Yes	Yes	Yes	Yes	Yes

Note: This table displays regression estimates using data from the validation survey conducted in May 2024 with an equal share of stockowners and non-owners (see Table A.2 for an overview of studies). The dependent variables in columns 1–4 are binary variables. "Non-owner" is a dummy for respondents without any exposure to the stock market. "active investing mental model" is a dummy for agreement with the active investing view. "Daily monitoring important" is a dummy for respondents selecting "daily" or "multiple times a day" when asked about how often it is necessary to monitor market conditions and check one's portfolio. "Quick reactions to market important" is a dummy for selecting "the key to stock market success is to react quickly to daily market changes." "Actively picking winner stocks important" is a dummy for selecting "the key to stock market success is to actively pick a few stocks that will increase in value and avoid stocks that will decrease in value." "Public info required to read" is a z-scored number of public information sources that respondents think one has to "regularly consult when participating in the stock market to make a return on one's portfolio." "Log weekly portfolio management time" is the log of the weekly hours that respondents think one would have to spend managing one's portfolio, measured on an interval scale from "less than 1 hour per week" to "more than 15 hours per week." All regressions include the following set of sociodemographic controls: age, gender, log household income, and indicators for having completed a college degree, working full-time, owning a house, having a mortgage, and having non-zero savings. "Financial literacy" is the number of correctly answered questions from the "Big 3" financial literacy inventory. Robust standard errors are shown in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Text production in AI-led interviews compared to open-ended survey questions

	Text length in characters		Increase in text production relative to survey questions	
	AI-led interview (1)	Open-ended survey questions (2)	Absolute increase (3)	Relative increase (4)
Panel A: Stock market interviews				
Opening question	187.3	166.8	20.5	12.3%
First question of topic 2	200.9	101.0	99.9	98.9%
First question of topic 3	231.3	137.0	94.2	68.8%
First question of topic 4	181.0	107.5	73.4	68.3%
Overall average	200.1	128.1	72.0	56.2%
Panel B: Tariff interviews				
Opening question	272.6	190.2	82.4	43.3%
First question of topic 2	257.2	112.6	144.6	128.4%
First question of topic 3	415.9	213.3	202.6	95.0%
First question of topic 4	288.9	190.0	98.9	52.0%
First question of topic 5	198.6	42.4	156.2	368.6%
Overall average	287.7	149.7	138.0	92.2%

Note: This table combines data from the stock market interview study and the corresponding benchmark survey (see Table A.2 for an overview of data collections) in Panel A. Panel B uses data from the tariff interview study and the corresponding benchmark survey. This table presents the average amount of text produced (in characters written per question) in response to questions in AI-led interviews (Column 1) and in response to standard open-ended survey questions (Column 2). Column 3 presents the absolute difference in text production (in characters) between AI-led interviews and open-ended survey questions for similar questions. Column 4 presents the relative increase in text production (in %) in AI-led interviews compared to an equivalent open-ended survey question. We present summary statistics separately for the opening question of each substantive interview topic. Note that the opening question of the interview is the opening question of Topic 1.

Table A.9: Response mode (voice vs typing) affects text production, but not survey time or richness of responses

	(1) Interview duration (in seconds)	(2) Text length (in characters)	(3) Assigned codes
Text input only	1,626.42	3,429.71	13.13
	(1,152.27)	(2,112.59)	(4.21)
Audio input used	1,633.18	7,275.01	14.31
	(1,025.22)	(4,994.80)	(4.33)
p-value of t-test for equality of means	0.952	0.000	0.009

Note: This table presents summary statistics for the duration of the interview (Column 1), text produced by interviewees during the interview (Column 2), and the number of unique codes assigned to the interview (Column 3) using data from the tariff interviews ($N = 385$; see Table A.2 for an overview of studies). The voice input feature was not available during the stock market interviews in 2023. Summary statistics are presented separately for the subsample of participants that never made use of the option to record their responses as a voice message ("text input only") and for the subsample who recorded at least one voice message ("audio input used"). 39% of interviewees used the voice message feature at least once. We present the mean and the standard deviation (in parentheses), as well as the p-value for a t-test of equality of means.

Table A.10: Interview codes are correlated with direct measurements of economic preferences, beliefs, and constraints

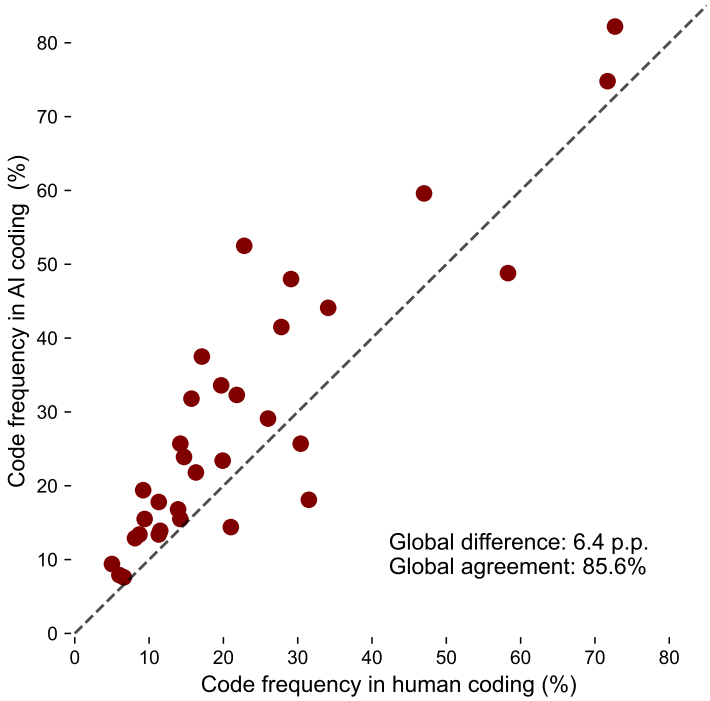
	(1) Risk-taking (Falk et al. 2018) (std.)	(2) Big 3 financial literacy (std.)	(3) Subjective belief: % chance of financial loss	(4) Log assets	(5) Age (in years)	(6) Homeowner (binary)
DOR: Dislike of risks	-0.547***					
	(0.101)					
LOK: Lack of knowledge		-0.259**				
		(0.113)				
FLL: Fear of losses			5.901**			
			(2.446)			
NFI: No funds				-0.924***		
				(0.236)		
TOR: Too old/retired					23.001***	
					(1.838)	
PTA: Prefer tangible assets						0.219**
						(0.102)
Constant	0.319***	0.188**	34.686***	10.499***	37.919***	0.044***
	(0.079)	(0.096)	(1.637)	(0.200)	(0.697)	(0.011)
N	381	381	381	354	381	381

Note: This table shows regression estimates using data from the stock market interview study (see Table A.2 for an overview of studies). The regressors are binary variables for whether a code (e.g., "Dislike of risks") was assigned to the participant based on the hand-coding of interview transcripts, or not. The dependent variable in column 1 is the willingness to take risks (Falk et al., 2018), which we standardize such that larger values denote higher willingness to take risks. The dependent variable in column 2 is the standardized number of correctly answered financial literacy questions from the "big 3" financial literacy inventory (Lusardi and Mitchell, 2011). The dependent variable in column 3 is the subjective belief about the probability of negative returns on the US stock market over the next 12 months. Column 4, 5 and 6 use the log assets, age (in years) and a binary dummy for whether participants are homeowners as dependent variables, respectively. Robust standard errors are shown in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

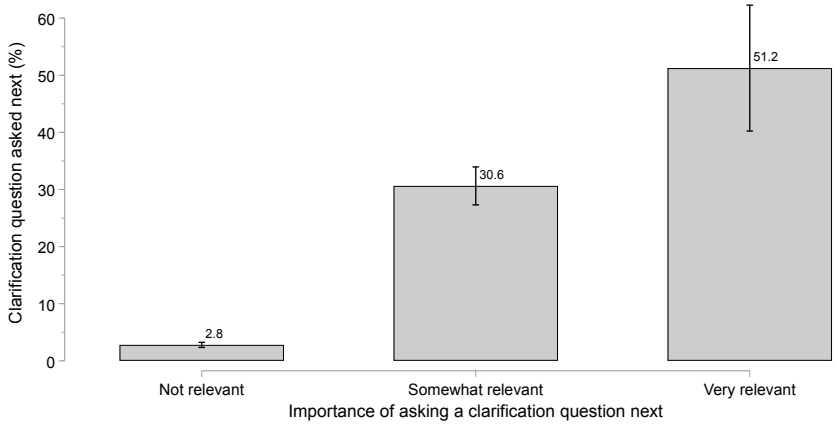
B Additional figures

Figure B.1: Validation of AI coding: Code frequencies derived from AI vs human coding across 12,192 decisions and 32 codes



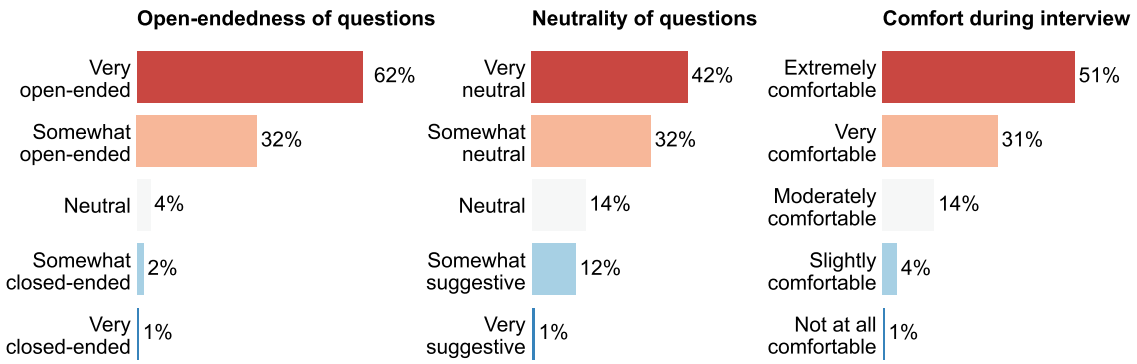
Note: This figure presents a scatterplot of code frequencies derived from human coding against those derived from AI coding, using the interview transcripts from the stock market interview study (see Table A.2 for a study overview). In particular, we construct a dummy variable at the participant level that takes value one if a given code was assigned at least once, and zero otherwise. Across all interview codes, the mean difference between AI coding and human coding is 6.4 percentage points. The global agreement between the AI and human coders is 85.6% across 12,192 interview-code pairs.

Figure B.2: AI interviewer asks clarification questions in response to ambiguous responses



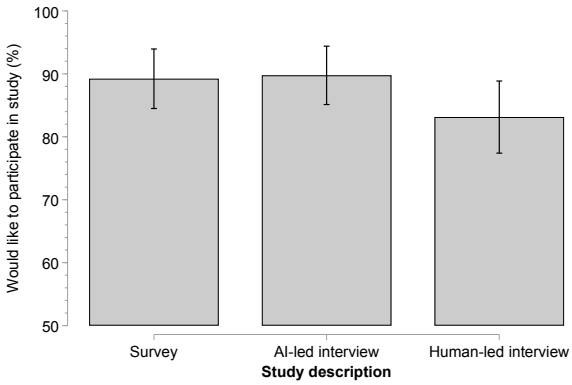
Note: This figure uses hand-coded data from our interviews. Participants' responses are rated based on the need for further clarification due to ambiguous statements ("Not relevant," "somewhat relevant," or "very relevant"). We then coded whether the *next* interview question is indeed a clarification question, or not. We plot the frequency of clarification questions as a function of the need for clarification. 95% confidence intervals are shown as vertical lines.

Figure B.3: Tariff interviews: Additional participant perceptions of questions



Note: This figure uses survey data from the tariffs interview study (see Table A.2 for an overview of studies). Each panel plots the distribution of responses to different Likert scale questions about the AI-led interview. "Open-endedness of questions" asked respondents to evaluate the open-endedness of the questions. "Neutrality of questions" asked respondents to evaluate the neutrality of the questions. "Comfort during interview" is participants' comfort in interacting with the AI interviewer during the interview.

Figure B.4: Demand for surveys compared to AI- and human-led interviews

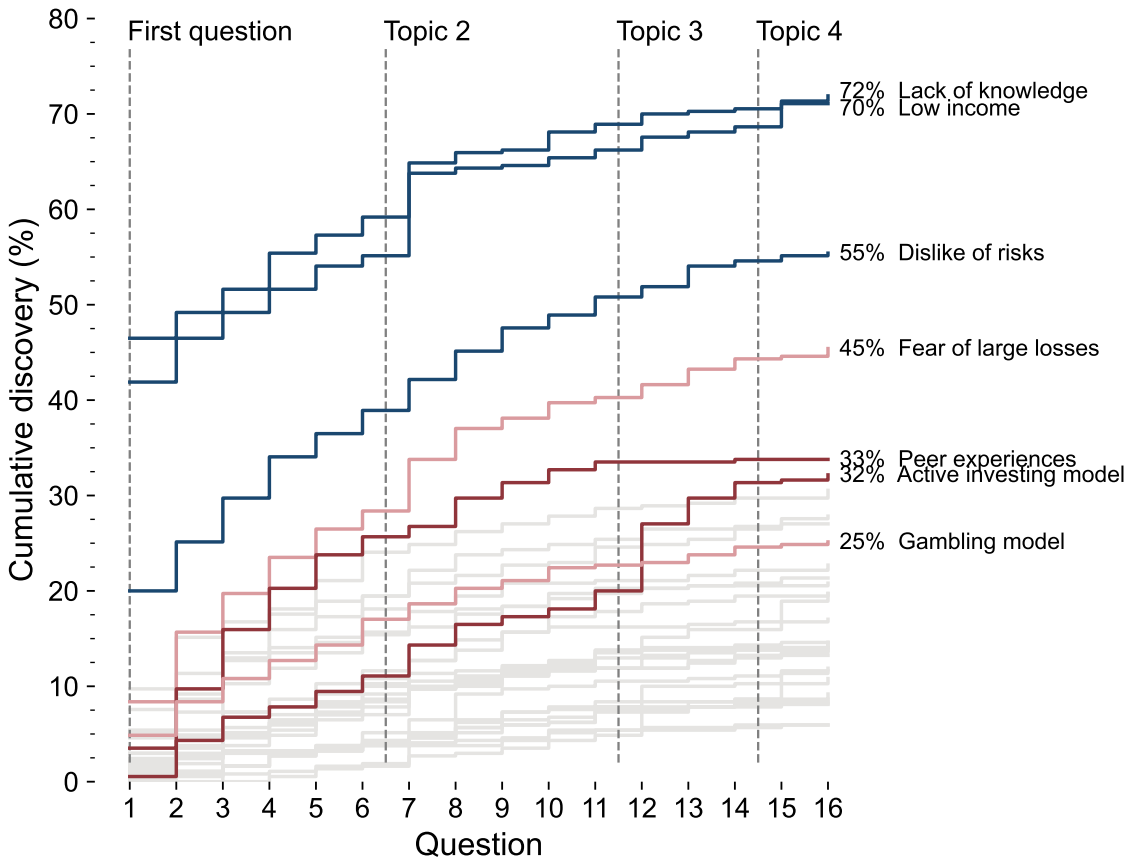


Note: This figure uses data from the first vignette in the selection experiment (see Table A.2 for a study overview). "Would like to participate in study (%)" is the share of respondents who would like to participate in a 40-minute study that was offered to them. The description of the study varied between respondents. The figure separately presents the demand for studies described as (i) a regular survey, (ii) a survey including a text-based interview with an AI, and (iii) a survey including a text-based interview with a human. 95% confidence intervals shown.

Figure B.5: Interview mode and code discovery: Code-level results

Note: Data from the tariff interviews where participants had the option to record a voice message and the option to type their responses with their keyboard (see Table A.2 for an overview of studies). This figure presents the share of respondents that can be assigned a specific code based on their interview transcript. "Text only" refers to the subset of respondents that exclusively responded to interview questions by typing their response. "Audio used" refers to the subset of respondents who used the option to answer interview questions by recording a voice message that is automatically transcribed at least once during the interview. Codes assigned by AI coding. We report code frequencies separately by subsamples.

Figure B.6: The timing of discoveries over the course of the stock market interviews



Note: This figure uses hand-coded interview transcripts from the stock market interview study. We plot the discovery process $d_c(t)$, i.e., the share of participants that can been assigned a specific code c based on all of their responses to the first t interview questions. Topic transitions are indicated as vertical lines. Each curve represents the discovery process for a separate code (see Figure 4). Selected codes are highlighted.

C Additional results

C.1 The timing of discoveries

Our findings from Section 5 that AI-led interviews can lead to the discovery of themes and recurring patterns that would be largely unobserved in text responses to a single open-ended survey question raise a natural follow-up questions about the timing of discovery: How long does it take to reveal a previously unobserved factor? What is the marginal return to asking *another* interview question? How does the discovery process differ across codes?

To address these questions, we leverage the fact that the interview transcripts are hand-coded at the response level. We derive the share $d_c(t)$ of respondents that can be assigned code c based on their cumulative responses to the first t interview questions, but not taking into account information that was revealed later in the interview. We refer to $d_c(t)$ as the *discovery process* for code c. Figure B.6 plots the discovery process of all codes that are assigned to at least 5% of respondents by the end of the interview.

For the top-of-mind codes *Lack of knowledge* and *Low income*, the first six probing questions capture most of the discoveries, with marginal returns diminishing beyond that point. In contrast, *Dislike of risks*, another top-of-mind response, shows a more gradual discovery process that only plateaus near the final questions. These patterns of gradual discovery are even more pronounced for the "high-gain" codes, such as *Gambling model* or *Active investing model*, where the marginal returns from further probing remain high throughout the interview. The discovery process for *Active investing model* even shows a steep rise after the third topic transition, where the interviewer introduces a "what if" scenario involving stock market investment. Similar jumps for other codes around the topic transitions demonstrate the value of a topic guide that explores the same underlying questions from multiple angles.

Overall, our findings underscore that the optimal interview length depends on the research questions. For researchers mainly interested in top-of-mind responses, a short interview with a few probing questions might be sufficient. For researchers interested in exploring more complex issues—such as narratives, mental models, and experiences that affect choices—the marginal returns to a longer interview are much higher. In our context, including 16 questions in total seemed to strike a good balance between maximizing discovery and maintaining respondent engagement throughout the interview.

C.2 Additional qualitative results: Stock market

Economic constraints Despite recruiting respondents with a minimum household income of $30,000 and a median financial assets of $18,000 (Table A.3), many interviewees start the interview by saying that they do not have the financial means to invest in the stock market. This belief is partly driven by misconceptions regarding the necessary minimum investment

amount. Several respondents assume that substantial upfront savings are required, either because they believe institutional rules dictate high entry thresholds for opening investment accounts or because they perceive that "good" stocks—those with low risk and high return—are prohibitively expensive, costing thousands of dollars per share: "…the most successful stocks have high buy prices and I think an onlooker such as myself would be enticed, yet unable to afford such luxuries." Other respondents argue that stock market participation requires the financial capacity to absorb losses and stay invested until returns are realized, which aligns with the broader mental model of the stock market as a casino: just as a gambler needs sufficient funds to endure a losing streak, respondents think that only those with substantial savings can navigate the market's volatility. For them, this suggests that the stock market is a domain reserved for the wealthy, who can afford to remain in the "game" long enough to see a return.

Experiences and social influences The perception of the stock market as excessively volatile is often shaped by respondents' direct or indirect experiences with investing. Many respondents point to the negative experiences of friends, family members, or coworkers who lost large amounts of money in the stock market, especially during the 2007–2008 financial crisis. This crisis looms large in respondents' memories. For many, the market is synonymous with a series of unpredictable crises and market crashes, which is essentially the only narrative about the market they recall from the news. Additionally, some respondents cite social influences rooted in their growing up in an environment where stock ownership was uncommon or viewed with skepticism. Some describe coming from families that emphasized financial caution, where stock market participation was discouraged as a reckless and risky activity. This socialization then results in identity-based arguments that the stock market is not for people "like myself" (Henkel and Zimpelmann, 2023).

D Additional details about AI-led interviews

This section contains additional details and discussions of the AI interviewer. Section D.2 contains a discussion of how our AI interviewer system can currently be customized and provides avenues for potential future extensions. Section D.1 contains details on the technical implementation, the parameter settings and prompts used for the AI interviewer. Section D.3 contains a discussion of considerations and practical advice for conducting interviews with an AI-interviewer.

D.1 Prompts

We conducted our interviews with the frontier model available at the time. In 2023, we use the `gpt-4-0613` and `gpt-4-0314` versions of the GPT-4 model family for the stock market interviews. In 2025, we use `gpt-4o-2024-08-06`. The stock market interviews made use of the probing, summary, and topic agents. The tariff interviews used the probing, topic and moderator agents of our modular framework (see Section 2 for a description of these agents).

We provide the complete prompt templates for our agents below. These templates include `{placeholders}` that are replaced programmatically to reflect the current state of the interview. For example, `{summary}` is replaced with the summary agent's current synopsis of the conversation on previous interview topics if the "summarization" option is activated. Otherwise, it is replaced with the full and non-summarized conversation (see Section 2 for a description of this optional feature). Similarly, the placeholder `{current_topic_history}` is replaced with the verbatim questions and answers belonging to the currently investigated interview topic. These messages are inserted into the prompt using the convention "Interviewer: ... Interviewee: ...".

For additional implementation details and model configurations, we refer to our open-source repository with setup instructions: `https://github.com/fchop/interviews`

D.1.1 Probing agent

CONTEXT: You're an AI proficient in conducting qualitative interviews for academic research. You conduct a qualitative interview with the goal of learning the interviewee's reasons for not investing in the stock market.

INPUTS:
A. Previous Conversation Summary: {summary}
B. Current Interview Topic: {current_topic}
C. Current Conversation: {current_topic_history}

TASK: Your task is to formulate the next probing question for the Current Conversation. The question should align with the Current Interview Topic, helping us to better understand and systematically explore why the interviewee is not participating in the stock market.

GENERAL GUIDELINES:
1. Open-endedness: Always craft open-ended questions ("how", "what", "why") that allow detailed and authentic responses without limiting the interviewee to "yes" or "no" answers.
2. Neutrality: Use questions that are unbiased and don't lead the interviewee towards a particular answer. Don't judge or comment on what was said. It's also crucial not to offer any financial advice.

3. Respect: Approach sensitive and personal topics with care. If the interviewee signals discomfort, respect their boundaries and move on.
4. Relevance: Prioritize themes central to the interviewee's stock market non-participation. Don't ask for overly specific examples, details, or experiences that are unlikely to reveal new insights.
5. Focus: Generally, avoid recaps. However, if revisiting earlier points, provide a concise reference for context. Ensure your probing question targets only one theme or aspect.

PROBING GUIDELINES:
1. Depth: Initial responses are often at a "surface" level (brief, generic, or lacking personal reflection). Follow up on promising themes hinting at depth and alignment with the research objective, exploring the interviewee's reasons, motivations, opinions, and beliefs.
2. Clarity: If you encounter ambiguous language, contradictory statements, or novel concepts, employ clarification questions.
3. Flexibility: Follow the interviewee's lead, but gently redirect if needed. Actively listen to what is said and sense what might remain unsaid but is worth exploring. Explore nuances when they emerge; if responses are repetitive or remain on the surface, pivot to areas not yet covered in depth.

YOUR RESPONSE:

D.1.2 Topic agent

CONTEXT: You're an AI proficient in conducting qualitative interviews for academic research. You're guiding a semi-structured qualitative interview about the interviewee's reasons for not investing in the stock market.

INPUTS:
A. Previous Conversation Summary: {summary}
B. Current Conversation: {current_topic_history}
C. Next Interview Topic: {next_interview_topic}

TASK: Introducing the Next Interview Topic from the interview plan by asking a transition question.

GUIDELINES:
1. Open-endedness: Always craft open-ended questions ("how", "what", "why") that allow detailed and authentic responses without limiting the interviewee to "yes" or "no" answers.
2. Natural transition: To make the transition to a new topic feel more natural and less abrupt, you may use elements from the Current Conversation and Previous Conversation Summary to provide context and a bridge from what has been discussed to what will be covered next.
3. Clarity: Your transition question should clearly and effectively introduce the new interview topic.

RESPONSE FORMAT: Your response should use the template below:
"'Question: "Insert your transition question here" "'

Remember to include "Question:" in your response. Start your response here:

D.1.3 Summary agent

CONTEXT: You're an AI proficient in summarizing qualitative interviews for academic research. You're overseeing the records of a semi-structured qualitative interview about the interviewee's reasons for not investing in the stock market.

INPUTS:
A. Interview Plan: {topics}
B. Previous Conversation Summary: {summary}
C. Current Topic: {current_topic}
D. Current Conversation: {current_topic_history}

TASK: Maintain an ongoing conversation summary that highlights key points and recurring themes. The goal is to ensure that future interviewers can continue exploring the reasons for non-participation without having to read the full interview transcripts.

GUIDELINES:
1. Relevance: Prioritize and represent information based on their relevance and significance to understanding the interviewee's reasons for not investing in the stock market.
2. Update the summary: Integrate the Current Conversation into the Previous Conversation Summary, ensuring a coherent and updated overview. Avoid adding redundant information.
3. Structure: Your summary should follow the interview's chronology, starting with the first topic. Allocate space in the summary based on relevance for the research objective, not just its recency.
4. Neutrality: Stay true to the interviewee's responses without adding your own interpretations of inferences.
5. Sensitive topics: Document notable emotional responses or discomfort, so subsequent interviewers are aware of sensitive areas.
6. Reasons: Keep an up-to-date overview of the interviewee's reasons for non-participation.

Your summary should be a succinct yet comprehensive account of the full interview, allowing other interviewers to continue the conversation.

RESPONSE FORMAT: Your response should use the template below:
"'Summary: "Insert your summary here" "'

Remember to include "Summary:" in your response. Start your response here:

D.1.4 Moderation agent

You are monitoring a conversation that is part of an in-depth interview. The interviewer asks questions and the interviewee replies. The interview should stay on topic. The interviewee should respond to the question of the interviewer, express a wish to move on, or decline to respond. The interviewee is also allowed to say that they don't know, do not understand the question, or express uncertainty. The interviewee's response might contain spelling and grammar mistakes. Here is the last part of the conversation.

Interviewer: '{last_question}.'
Interviewee: '{user_answer}.'

That is the end of the conversation. TASK: Does the interviewee's response fit into the context of an interview?

Please answer only with 'yes' or 'no'

D.2 Modifications and extensions of the AI interviewer system

Our modular AI interviewer system can be tailored to a single-agent or a multi-agent workflow that allows researchers to add or subtract "agents" as needed. This lightweight structure ensures low latency and makes modifications to the system straightforward.

Endogenous topic transitions Our modular framework supports endogenous topic transitions where the AI interviewer has the choice to ask either more or fewer questions than indicated in the topic guide. To do so, researchers should deactivate the optional topic agent and the optional summarization agent. In addition, the full content of the topic guide should be added to the probing agent's prompt to inform the agent about the intended interview structure. This enables the probing agent to endogenously decide if and how topic transitions should be managed. For details about the parameters and how to deactivate agents, see the instructions in our open-source repository: `https://github.com/fchop/interviews`.

We conducted our interviews with a topic agent because we prefer precise control over the number of questions per topic. This will be relevant whenever (i) researchers would like

interviews to be comparable across participants and (ii) need to communicate the approximate interview duration to participants at the recruitment stage.

Voice and facial recognition Our open-source application supports both audio input and text input from interviewees. Audio input is transcribed to text using OpenAI's Whisper API, which requires about one to two seconds to transcribe even longer audio messages. 40% of our participants used this feature. See Section 5.3 for our descriptive results on how this boosts text production. However, one could be interested in using raw audio or video recordings as inputs for multi-modal models. Of course, this raises its own data protection and privacy issues, and may negatively affect recruitment if participants feel uncomfortable with having their voice or image processed. We therefore expect that voice-input-to-text—without actually storing the underlying voice message—to be the preferred approach in many applications.

In principle, implementing voice and facial recognition with newer models, like OpenAI's GPT-4o, which are trained on datasets including text, audio, and video, is technically feasible. The key bottleneck for this is the inference speed of the multi-modal model, which is outside the researcher's control as it depends primarily on the technical infrastructure of the company hosting the model. The potential benefit of such an extension is that it makes it easier to interview "hard-to-reach" groups, although our voice-to-text input method makes the interview setup accessible to people who might otherwise find it difficult to express their words by typing. Moreover, nuances embedded in natural speech might make it easier to detect tension or uncertainty (Graeber et al., 2024c).

Multi-agent probing agent Another natural extension is to replace the probing agent with a set of more specialized agents, similar to ensemble techniques in machine learning (Dietterich, 2000). For example, probing questions can be conceptually disaggregated into three distinct categories: *clarification* requests, follow-up questions aimed at increasing *depth* in responses, and questions aimed at expanding the *breadth* of the conversation. An alternative design would thus consist of three separate language models tasked with proposing a clarification, depth, and breadth question, respectively. A fourth language model then chooses one of the questions. While this may yield potential returns of higher question quality, we expect these benefits to be more modest in practice, and that these benefits have to be weighted against the potential increase in system complexity. We note, however, that such so-called "agentic workflows" (Park et al., 2023) have increasingly become the norm in industry applications.

Fine-tuning the foundational model In our applications, we did not fine-tune a foundational model, instead relying on the off-the-shelf version. However, the quality of probing questions could potentially be improved by fine-tuning domain-general language models on training data consisting of high-quality interview transcripts from human subjects (Ziegler et al., 2019). While this raises ethical concerns vis-à-vis the human subjects represented in the training data, fine-tuning might allow researchers to encode the difficult-to-describe implicit notion of what

constitutes an effective probing question. However, our experience suggests that the returns from model fine-tuning might be limited, as frontier models already achieve quite robust performance.

D.3 Guide for AI-led interviews

In this section, we provide a short step-by-step guide with key considerations for researchers who are interested in conducting AI-led interviews using our open-source platform.[1]

Ask whether AI-led interviews are the right tool for you The first step is to evaluate whether an AI-led interview is the best choice for your particular setting and research question. While AI-led interviews allow for scale and offer high-quality interview data at low cost, traditional interviews led by humans might sometimes still be a better choice in some situations. For instance, given that current LLMs display worse context sensitivity and ability to detect emotional nuances than humans (Bano et al., 2023), interviews where these factors are likely to be important might be better served by a human interviewer. Furthermore, for sensitive topics, and in situations with heightened privacy concerns, a human interviewer might also be preferred.

Selecting agents If you have decided that AI-led interviews are the right approach for you, you first need to decide which agents to include in your interview setting. By default, our AI interviewer consists of four underlying agents: a probing agent, a summary agent, a topic agent, and a moderator agent. However, the system is modular and the only mandatory agent is the probing agent. How many agents you need depends on your research question and the purpose of the interview. If the point of the interview is to only clarify potentially ambiguous responses from an open-ended survey question that is part of a larger survey, a single-agent workflow with only the probing agent (and the moderator agent) activated might be sufficient. For a longer interview, including more agents in the set-up can be advantageous for greater control and better performance. For example, by using the summary agent to compile analytic notes for the other agents. For a semi-structured interview in particular, a topic transition agent can improve the flow and help make sure that all interviews cover roughly similar ground. Furthermore, for very long interviews beyond the 30-60 minute interviews we focus on, the fixed context length of the LLM agent might eventually be exceeded, making a summary agent necessary to preserve the essential points from the conversation history. However, since the latest generation of LLMs have very long context windows, we make it easy to deactivate the summary agent in our open-source software. Lastly, including a moderator agent can be beneficial for security reasons, even for very short conversations.

Adapting the topic guide The next step is to adjust the topic guide to fit your specific research context and, if needed, modify the agent prompts with custom instructions. While the prompts in our open-source software are designed to be easily portable across settings by avoiding application-specific instructions, there might be a small number of changes needed before

[1]The platform is available at the following link: `https://github.com/fchop/interviews`.

they can be taken to a new setting. For example, when discussing a recent political event, it might be necessary to provide a short description of these events in the prompt if the event is outside the knowledge cutoff of the model. More broadly, it is always necessary to change the context provided to the agents at the beginning of each prompt. For example, in our stock market interviews, we specify the following to the probing agent: "You conduct a qualitative interview with the goal of learning the interviewee's reasons for not investing in the stock market." Interviews in other settings must naturally specify a different goal or interview topic. Researchers who want to conduct semi-structured interviews also need to adjust the topic guide, i.e., provide a list of topics that should be covered by the AI interviewer. For economists unfamiliar with how to structure a topic guide, we recommend consulting Knott et al. (2022) for a short primer and Arthur and Nazroo (2003) for a more in-depth guide. See also Table A.1 for how we designed the topic guides in our applications.

Piloting We recommend that researchers conduct a small number of mini interview pilots with 5 to 10 participants each to gather participant feedback before scaling up the data collection. This iterative approach allows researchers to spot early on whether they would like to make any custom changes and, more importantly, get a feeling for how people approach the interview topic in a real world interview. This, in turn, will often lead to minor revisions of the topic guide. These pilots can be efficiently conducted within a few days.

Data analysis After the data collection, the researchers need to decide how to analyze the data. The first step is usually to discover prominent themes in the data. The traditional approach to qualitative discovery is to read through all interview transcripts in detail. This process can be time-consuming and may become increasingly impractical when conducting interviews at scale. While LLMs can certainly help in the discovery process, they also face some potential limitations, including a strict reliance on well-crafted prompts to work well, the potential to hallucinate (finding patterns without a clear justification in the data), and difficulties in interpreting subjective expressions, complex meanings, and emotional nuances (Barros et al., 2024). It is therefore important to use AI tools as an aid rather than a replacement for human analysis (Mathis et al., 2024; Carius and Teixeira, 2024; Paoli, 2024), making sure that LLM-assisted discoveries are carefully monitored and validated by humans to mitigate against hallucination and systematic model biases (Carlson and Dell, 2025). In Section 2.3.1, we provide a scalable method for machine-assisted discovery.

Coding of the data For a quantitative research project, the final step of the process consists of systematically coding interview transcripts based on a coding manual. Several studies demonstrate that LLMs can code qualitative data with performance close to that of humans (Bursztyn et al., 2023a; Braghieri et al., 2024; Link et al., 2024; Gilardi et al., 2023), making the process in principle scalable and more cost-efficient compared to traditional human coding. However, the quality of AI coding depends on both (i) the quality of the prompts and (ii) the

complexity of the task. We therefore recommend to conduct human coding for a sufficiently large random subset of the data to validate the quality of LLM coding. If the correlation between human and LLM coding is not satisfactory, the prompts should be refined to improve performance.

E Additional details on the coding of interviews

E.1 Human coding

This section contains additional information about the hand-coding of interview transcripts. Four trained research assistants from the University of Copenhagen and the NHH Norwegian School with field experience as interviewers or experience working in qualitative research projects, were given the task of hand-coding the interviews according to various coding schemes.

Training Research assistants read a handbook chapter on interviewing methods (Legard et al., 2003) and were provided with detailed coding manuals that included the definition of each variable along with several examples of what type of questions or answers would fall into the different categories. Research assistants received extensive training before starting with the task in the form of multiple rounds of double and triple coding, followed by group discussions about how to reconcile coding differences until a joint interpretative understanding of all codes was reached. The probability of agreement for a randomly selected code is 83% across research assistants.

Thematic coding We randomly assigned interviews to research assistants. For each code, research assistants indicated their level of confidence in the code (high vs low). Research assistants assigned codes to individual responses to interview questions. A code was assigned to the kth response if the conversation up until that point included sufficient unambiguous evidence to warrant assigning the code. If in doubt, research assistants were instructed to wait until the code could be unambiguously assigned at a later point during the interview when more evidence had accumulated. Each response could be assigned multiple codes.

Stock market interviews We conducted two separate rounds of hand-coding for all stock market interview transcripts, which include about 12,000 questions and responses. The first round of hand-coding focused on the nature and properties of questions asked by the AI interviewer and the subsequent responses (e.g. whether questions are open-ended or responses display signs of discomfort; see Section 3.2). The second round of hand-coding, conducted after the first round of coding, was based on the coding manual derived from the inductive thematic analysis and focused on coding participants' reasons for not participating in equity markets. A direct benefit of conducting two rounds of coding is that research assistants had already immersed themselves in the interview transcripts before starting with the more demanding task of coding respondents' reasons for not participating in the stock market. When we discussed the corresponding coding manual for the first time, the research assistants already had a robust understanding of many codes and could immediately recognize the concepts from interviewers they had read previously. This ensured a high shared conceptual understanding of corresponding codes. The research assistants were also told that creating a coding scheme is an iterative process and were encouraged to

suggest potential changes to the coding scheme for stock market non-participation reasons based on their readings of the interviews.

Tariff interviews For tariff interviews, 200 out of 384 interview transcripts were randomly selected and hand-coded for the analyses presented in Section 3. This amounts to about 6,000 questions and responses. This annotation was carried out after the annotation of the stock market interviews.

Table E.1: Coding manual for the first round of hand-coding

Category	*Description*	*Levels*
	Questions by the AI interviewer	
Open-ended	Assessment of whether questions allow for detailed responses or are constrained to short, direct answers.	4 (open, closed, composite, other)
Leading	Determination if questions suggest a specific answer or allow for open-ended responses.	2 (leading, non-leading)
Relevance	Evaluation of how closely questions align with the main topic or goal of the interview.	3 (not at all, somewhat, very)
Redundancy	Evaluation of whether questions introduce new elements or repeat previously covered points.	2 (not redundant, redundant)
Clarification	Classification of questions based on their role in clarifying ambiguities or seeking additional details.	3 (non-language, language, other)
Recall	Analysis of whether questions refer back to earlier parts of the conversation for context or continuity.	4 (no, distant, close, incorrect)
Quality	Overall subjective assessment of question quality.	2 (low, high)
	Respondent Answers	
Engagement	Evaluation of the depth and thoughtfulness of responses.	3 (no, low, high)
Discomfort	Analysis of whether responses indicate unease or discomfort related to the question's nature.	2 (discomfort, no discomfort)
Annoyance	Assessment of whether responses exhibit annoyance or frustration towards the question or interviewer.	2 (annoyance, no annoyance)
Clarification needed	Determination of the need for additional probing or clarification based on the respondent's answers.	3 (not, somewhat, very relevant)

Note: This table presents the coding manual for the codes and variables discussed in Section 3 on effort, interview experience, and the AI interviewer's alignment with its instructions. A team of trained research assistants coded all non-deterministic questions and responses. Codes in the top part of the table are only assigned to questions, while codes in the bottom part are only assigned to respondents. Codes are assigned based on the full conversation history up to that point, but not taking into account future questions and answers.

E.2 AI coding

We use the following prompt template for AI coding. The `{placeholders}` in the template are replaced with information from the coding manual, and the interview transcript is appended at the end of the prompt.

Context: You are an expert in coding qualitative interviews with a strong background in sociology and qualitative research methods. You are given an interview transcript containing numbered question-answer pairs exploring why the respondent does not participate in the stock market.

Task: Your task is to decide whether there is clear evidence of the code 'code' using a systematic three-step approach. Follow the steps below carefully.

Step 1: Extract Evidence
*- Identify the **strongest evidence** from the interview transcript that directly relates to the definition of code.*
- Reference specific answers from the transcript (e.g., question 3).
- Include supplementary evidence only if it provides essential context or clarification.
- If no relevant evidence exists in the transcript, explicitly state this.
*- **Do not introduce any information not present in the transcript.***
Step 2: Evaluate Evidence
*- Determine if the extracted evidence is **unambiguous** according to these criteria: **Assign '1' if:** - The evidence directly matches the definition of code without making further assumptions.*
- The respondent explicitly states personal experiences or beliefs that align with code.

***Assign '0' if:** - The evidence is ambiguous or fits another code better.*
- Conflicting evidence weakens the case for code.

*- **Handling Overlaps with Other Codes:****
- If the evidence aligns better with another code, exclude it from code and justify the alternative classification.

Step 3: Justify Your Decision
- Provide a clear justification for your classification.
- Reference the extracted evidence explicitly.
- Explain why the evidence meets or does not meet the criteria for code.
- If assigning '0', clarify why the evidence is ambigious or better fits another code.

Coding Guidelines:
*1. **Evaluate All Responses:** Review every question-answer pair for full context.*
*2. **Consider Related Codes:** Before confirming, check if the evidence aligns more closely with another code.*
*3. **Combine Relevant Evidence:** Include all answers that collectively support or refute the code.*
*4. **Conservative Approach:** Assign '1' only if the evidence unambiguously meets the definition.*
*5. **Context Matters:** Supplementary answers can provide context but cannot create evidence.*
*6. **Use Examples for Guidance Only:** Refer to the examples to understand the definitions, but do not rely on them when extracting evidence. Base your analysis solely on the interview transcript without introducing information from the examples.*

Full Coding Scheme for Reference: {list of all codes}
Specific Guidance on code: {definition}
Examples (for reference only): {examples}
Response Format:
Return the following structured response and nothing else:

{ "primary_evidence": "[The specific answer from the below interview transcript that provides the strongest evidence.]",
"supplementary_evidence": "[Additional context from the interview transcript, or 'None'.]",
"classification": [1 or 0],
"justification": "[A brief explanation that justifies the classification based on the extracted evidence.]" }

Interview Transcript: {transcript}

F Experimental instructions

This section contains the core experimental instructions for our studies. Text in *italics* is not part of the instructions. Table A.2 provides an overview of the data collections.

F.1 Stock market studies

F.1.1 Screener survey (August 2023)

The next question is about the following problem. In questionnaires like ours, sometimes there are participants who do not carefully read the questions and just quickly click through the survey. This means that there are a lot of random answers which compromise the results of research studies. To show that you read our questions carefully, please choose both "Extremely interested" and "Not at all interested" as your answer in the below question.

Given the above, how interested are you in sports? [Extremely interested, Very interested, A little bit interested, Almost not interested, Not at all interested]

Only respondents who select (a) and (e) pass the attention check.

Please explain: What is your opinion about the yearly switch to daylight saving time? Do you like or dislike it? Please use about 15-30 words.

Only respondents who write at least 20 words are eligible to participate in an interview.

What was your family's gross household income in 2022 in US dollars? [Dropdown menu]

1. How do you currently save or invest your money? Please check all that apply.

a) I am currently unable to save money
b) Savings or checking account
c) Certificates of Deposit (CDs)
d) Bonds or bond funds
e) Money market accounts
f) Individual stocks or stock mutual funds
g) Real estate or property
h) Retirement accounts (e.g. 401k, IRA)
i) Other (please specify): _____

2. Do you currently own any individual stocks or stock mutual funds?

a) Yes, I own individual stocks.
b) Yes, I own stock mutual funds.
c) Yes, I own both individual stocks and stock mutual funds.
d) No, I do not own any individual stocks or stock mutual funds.

3. Do you plan to buy individual stocks or stock mutual funds over the next three months? (Yes / No)

4. Do you have a financial advisor who manages your investments for you? (Yes / No)

Only if (d) was selected in Question 2:

5. Do you own any individual stocks or stock mutual funds indirectly, such as through pension savings, retirement accounts, or other investment vehicles?

a) Yes, I own stocks or stock mutual funds indirectly.
b) No, I do not own any stocks or stock mutual funds indirectly.

Only respondents who selected (a) and not (e) in Question 1; (d) in Question 2; "No" in Question 3 and 4; and (b) in Question 5 were eligible to participate in an interview on stock market non-participation.

F.1.2 Stock market interview survey (August/September 2023)

[Note: Pre-interview characteristics]

How often do you typically use AI tools such as ChatGPT, Bard or Bing per week? [Several times a day, About once a day, A few times a week, Once a week, Less than once a week, Never]

(New survey page)

Please tell us about yourself.

What is your age? [Dropdown menu]

What is your gender? [Male, Female]

Which category best describes your highest level of education? [Dropdown menu]

What is your current employment status? [Dropdown menu]

[Note: Introducing the interview]

Next up: The Interview

You will now participate in an interview with an AI chatbot that will ask you questions, similar to how a human interviewer would. The topic of the interview is stock market participation. The AI chatbot is informed that you indicated to us in a previous survey that you do not currently own any stocks or stock mutual funds. The interview will take about 20 minutes.

How it works

The interview uses a chat interface, much like texting on your phone. When a question appears, you can type your response. After your response, it takes between 5 to 10 seconds for the next question to appear. Take your time in crafting your responses - there is no rush. And don't worry, there are no right or wrong answers.

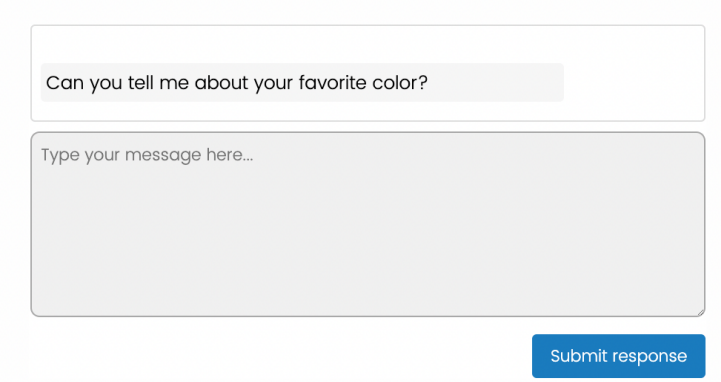
Please note

Your responses are entirely anonymous and will be used solely for research purposes. Please do not reveal any personally identifiable information such as your name. The AI chatbot is instructed not to give financial advice or explain financial concepts. Kindly avoid such queries.

Try it out

Below is an example of the chat interface. Please try it out before you proceed.

Figure F.1: Screenshot of the training chat interface



Note: This figure provides a screenshot of the chat interface that respondents could interact with to familiarize themselves with it. Respondents were asked up to three questions about their favorite color before the "Submit response" button was deactivated. Respondents are then encouraged to proceed to the next page.

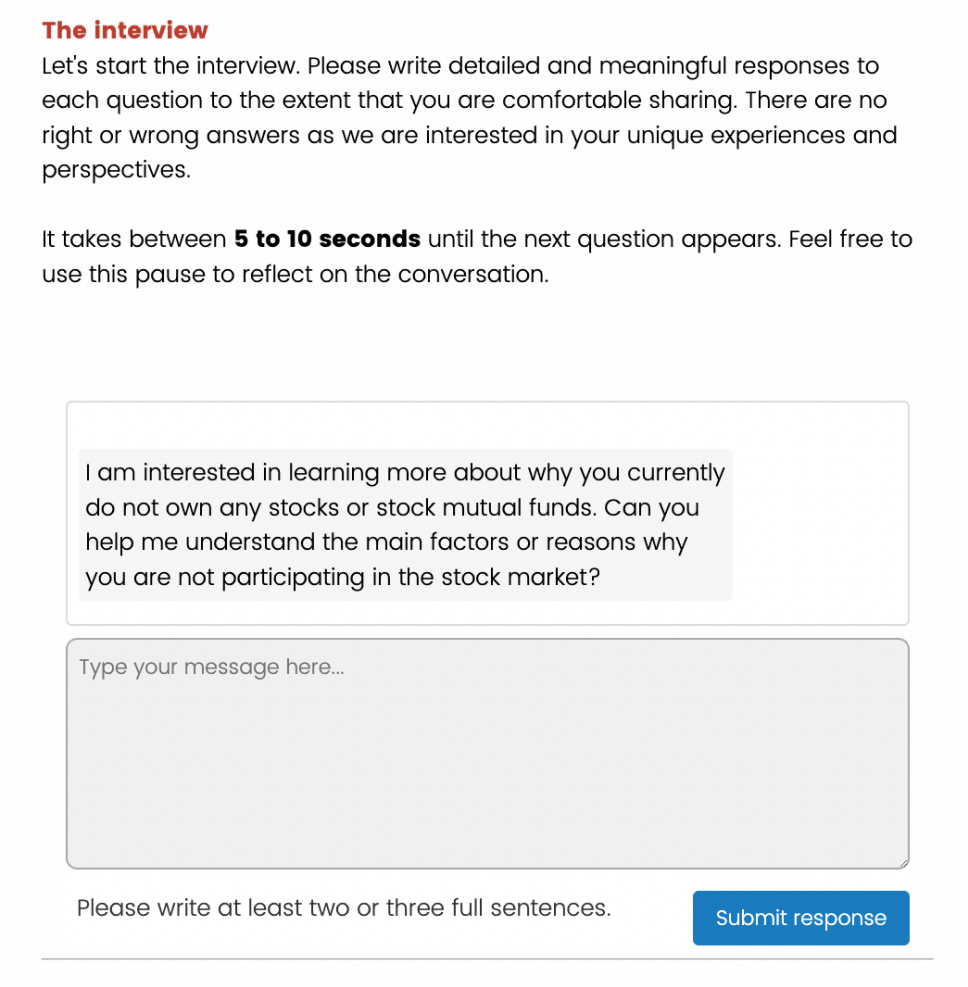
[Note: Interview block]

To ensure that the interview is as meaningful as possible, it's important that your responses provide sufficient detail. This means going beyond short statements, and diving into the "why" and "how". We are interested in your unique perspective and voice.

So, as we move forward, please aim to write at least two to three full sentences in your answers. This will provide enough depth and context for a rich and meaningful conversation. Of course, if you don't have an opinion on a topic or you're unsure, it's completely fine to express that too, with responses such as "I don't know".

(New survey page)

Figure F.2: Interview



Note: This figure provides a screenshot of the survey page that was used to conduct the stock market interviews.

(New survey page)

Here is a summary of the previous interview. *[Text box with a summary of the interview]*

Does the above summary accurately represent your views expressed in the interview? [Yes, No]

(Only if "No" was selected in the previous question:)

What is inaccurate in the above summary? *[Text box for open-ended responses]*

(New survey page)

Thank you for completing the interview with the AI chatbot!

Before this study ends, we have some traditional survey questions relating to your financial behavior and beliefs as well as a few demographic questions that we want to ask you.

[Note: Multiple-choice questions]

What are your main reasons for not participating in the stock market? Please check all that apply.

a) I don't know how to get started
b) Wealth too small to invest
c) Only few years until retirement
d) Need cash on hand / liquidity
e) Risk of losing a substantial part of investments
f) Uncertainty of returns
g) Don't trust financial professionals
h) Stock market is too complex
i) Setting up an investment account is too costly
j) Managing a portfolio is too costly (time, effort, money)
k) Don't like to think about my finances
l) Lack of knowledge
m) Negative experiences with the stock market
n) Stock market returns are not high enough
o) My labor income is too uncertain
p) Family, friends, or coworkers advise against stocks
q) Prefer more tangible assets
r) Intended to invest in stocks but never got around to it
s) None of the above

Except for "None of the above", we randomize the order in which items are presented.

[Note: Interview experience module]

How would you rate your overall experience with the interview conducted by the AI chatbot? [Extremely positive, Very positive, Somewhat positive, Neither positive nor negative, Somewhat negative, Very negative, Extremely negative]

How natural did the conversation with the AI chatbot feel? [Extremely natural, Very natural, Somewhat natural, Neutral, Somewhat unnatural, Very unnatural, Extremely unnatural]

If you were to participate in a future study involving a qualitative interview conducted through a chat interface, would you prefer texting with the same AI chatbot or an actual human interviewer? [Strongly prefer AI chatbot, Moderately prefer AI chatbot, Slightly prefer AI chatbot, No preference, Slightly prefer human interviewer, Moderately prefer human interviewer, Strongly prefer human interviewer]

Would you be interested in participating in an interview with an AI chatbot again? [Yes, No]

[Note: Post-interview module]

In this question we present eight different scenarios for US stock market returns over the next 12 months.

Please let us know how likely you think it is that each scenario will occur. Please type in the number to indicate the probability, in percent, that you attach to each scenario. The probabilities of the eight scenarios have to sum up to 100%.

The US stock market return over the next 12 months will be…

a) More than 30% : _____
b) Between 20% and 30% : _____
c) Between 10% and 20% : _____
d) Between 0% and 10% : _____
e) Between -10% and 0% : _____
f) Between -20% and -10% : _____
g) Between -30% and -20% : _____
h) Less than -30% : _____

Assume that your household's car broke down and the repair costs $1,000. How difficult would it be for your household to take out a loan to finance this repair? [Very easy, Easy, Neither easy nor difficult, Difficult, Very difficult]

In case of an unexpected decline in income or increase in expenses, does your household have at least two months of income available in cash, bank accounts, or easily accessible funds? [Yes, No]

(New survey page)

What is the current value of all financial assets (cash, savings accounts, bonds, stocks, etc.) that your household owns? Please include holdings through retirement accounts.
[Dropdown menu]

What is the current value of all non-mortgage debt (credit card, student debt, medical debt, auto loans, etc.) that your household owes?
[Dropdown menu]

Do you own or rent your current main residence? [Own, Rent, Other]

Do you have any mortgage debt outstanding against your household's main residence? [Yes, No]

(New survey page)

(Only shown if the respondent has outstanding mortgage debt:)

What is the current value of all mortgage debt (incl. home-equity based borrowing, HELOCs, etc.) that is outstanding against your household's main residence?
[Dropdown menu]

(Only shown for homeowners:)

What do you think is the current value that your household's main residence would sell for?
[Dropdown menu]

Suppose you had $100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow? [More than $102, Exactly $102, Less than $102, Don't know]

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, with the money in this account, would you be able to buy... [More than today, Exactly the same as today, Less than today, Don't know]

Do you think the following statement is true or false? Buying a single company stock usually provides a safer return than a stock mutual fund. [True, False, Don't know]

Which of the following statements is correct? If somebody buys the stock of firm B in the stock market, then... [He owns a part of firm B., He has lent money to firm B., He is liable for firm B's debts., None of the above statements are correct., Do not know.]

In general, how willing or unwilling are you to take risks?
[11-point scale from "0 (completely unwilling to take risks)" to "10 (completely willing to take risks)"]

How willing are you to give up something that is beneficial for you today in order to benefit more from that in the future?
[11-point scale from "0 (completely unwilling to do so)" to "10 (completely willing to do so)"]

Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people? [You can't be too careful, Most people can be trusted]

[Note: Sociodemographics]

Which of the following best describes your race or ethnicity? [Dropdown menu]

Are you of Hispanic, Latino, or Spanish origin? [Yes, No]

How many people, including yourself, currently live in your household? [Dropdown menu]

How many children do you have? [Dropdown menu]

What is your region of residence? [Dropdown menu]

In which state do you currently reside? [Dropdown menu]

(New survey page)

Thank you for participating in our survey!

If there are any remarks that you would like to make or clarifications that you would like to obtain, please do let us know by writing them into the field below. *[Text entry field]*

F.1.3 Stock market benchmark survey (November 2024)

We conducted a screener survey identical to the screener survey from August 2023. Eligibility to participate in the benchmark survey was then determined based on identical criteria as in the main survey from August 2023.

In a previous survey, you told us that you are currently not participating in the stock market. That is, you do not own any stocks or stock mutual funds. On the next pages, we will ask you six open-ended questions about your reasons for not participating in the stock market.

It's important that your responses provide sufficient detail. Please aim to write at least two to three sentences in your answers. If you don't have an opinion on a topic or you're unsure, you can express that with responses such as "I don't know".

There are no right or wrong answers; we are simply interested in your unique experiences and perspectives.

(New survey page)

We are interested in learning more about why you currently do not own any stocks or stock mutual funds. Can you help us understand the main factors or reasons why you are not participating in the stock market? [Text entry box]

(New survey page)

What barriers or challenges prevent you from participating in the stock market? [Text entry box]

(New survey page)

Imagine you invest in the stock market. What actions would you take, and what do you think would help you succeed? [Text entry box]

(New survey page)

What conditions or changes would need to happen for you to consider investing in the stock market? [Text entry box]

(New survey page)

As we conclude, are there any perspectives or information you feel haven't been addressed that you'd like to share? [Text entry box]

(New survey page)

Reflecting on our conversation, what would you identify as the main reason you're not participating in the stock market? [Text entry box]

F.1.4 Long-term interview follow-up survey (May 2024)

We include a standard attention check after the consent form and then elicit basic demographic characteristics: age, gender, education, employment, and income. We measure these as in our main survey. We do not replicate these instructions.

Demand for insurance against losses

An opportunity to get exposure to the US stock market

We will now give you the opportunity to decide between investing $1,000 in the US stock market for 12 months or getting $1,000 for sure in 12 months.

Loss protection

Loss protection allows you to shield your stock market investments from losses. How much you are shielded depends on the size of the loss protection. **The higher the loss protection, the less money you have at risk**. For example:

- With 0% loss protection, all your money is at risk. You can lose $1,000 of the initial $1,000 investment.
- With 50% loss protection, only half your money is at risk. You can maximally lose $500 of the initial $1,000 investment.
- With 100% loss protection, none of your money is at risk. You cannot lose a single dollar of your initial $1,000 investment.

If the stock market increases, you still get 100% of the upside no matter the level of your loss protection.

How much loss protection do you require to put your money at risk?

You will repeatedly choose between two options that involve payments to your Prolific account in **12 months**.

Both options are **worth $1,000 today**. The two options differ in how their value might change over the next 12 months:

- **Option A:** Will have a guaranteed return of **0%** (and thus be worth $1,000 in 12 months).
- **Option B**: Will provide the same return as the US stock market over the next 12 months (as measured by the S&P 500 index) with varying levels of loss protection.

(continued)

How do payments work?

1 out of every 200 survey participants will be randomly selected. If you are among the selected participants, we will randomly select one of the choices you make below and implement it.

- If Option A is implemented, we will transfer a $1,000 bonus to your Prolific account in 12 months.
- If Option B is implemented, we will transfer $1,000 adjusted for the return of the US stock market over the next 12 months and the amount of loss protection. We will also send you a monthly update about the value of your investment.

	Which option do you prefer?		
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **0%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **10%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **20%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **30%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **40%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **50%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **60%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **70%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **80%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **90%** loss protection.
$1,000 with a guaranteed 0% return.	○	○	$1,000 invested in the stock market with **100%** loss protection.

→

Willingness to pay for an investment course

How much do you value a stock market investment course?

MasterClass is an online education platform known for its wide range of professionally made courses with top experts from various fields.

They've recently introduced a four-part series aimed at helping **beginners understand the basics of stock market investments** and **learn how to successfully navigate the market**. As one of the instructors puts it, "Wall Street makes things very complicated, but it doesn't have to be that way."

How are choices made?

You will repeatedly choose between two options:

- **Option A:** You get free 3-month access to the "Mastering the Markets" stock market investment course from MasterClass.
- **Option B**: Varying amount of bonus payments to your Prolific account.

1 out of every 25 survey participants will be randomly selected. If you are among the selected participants, we will randomly select one of the choices you make below and implement it.

▸ Click **here** if you would like to know how you would receive access.

(continued)

	Which option do you prefer?		
Stock market investment course	○	○	**$0** bonus
Stock market investment course	○	○	**$2** bonus
Stock market investment course	○	○	**$4** bonus
Stock market investment course	○	○	**$6** bonus
Stock market investment course	○	○	**$8** bonus
Stock market investment course	○	○	**$10** bonus
Stock market investment course	○	○	**$12** bonus
Stock market investment course	○	○	**$14** bonus
Stock market investment course	○	○	**$16** bonus
Stock market investment course	○	○	**$18** bonus
Stock market investment course	○	○	**$20** bonus

→

Belief about the performance of actively managed funds

Your beliefs about the performance of actively managed funds

A stock market fund pools money from multiple investors to invest in a diversified portfolio of stocks.

It is common to make a distinction between active funds and passive funds:

- Active funds are managed by professional fund managers who actively trade stocks and aim to outperform the market by **picking superior stocks and monitoring market conditions**.
- Passive funds try to obtain the average market return without actively picking stocks or taking market conditions into account when buying and selling.

Every year a large investment research company measures how many percent of active funds were able to outperform their passive counterpart (after fees).

How many percent of U.S. active funds do you think outperformed their passive counterpart in 2023? If your answer is within +/- 2 percentage points of the actual answer, you will receive a **$2 bonus**.

I think [ ]% of active funds outperformed their passive counterpart.

Note: We will tell you the actual answer and whether you will receive the $2 bonus at the end of the survey.

→

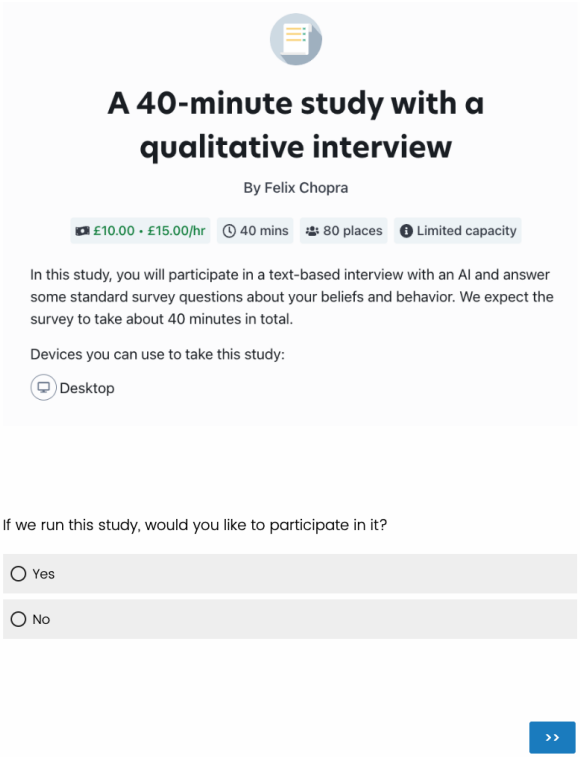
F.1.5 Selection experiment (November 2023)

We elicit similar sociodemographic variables as in the Screener Survey (August 2023). We do not replicate these instructions here.

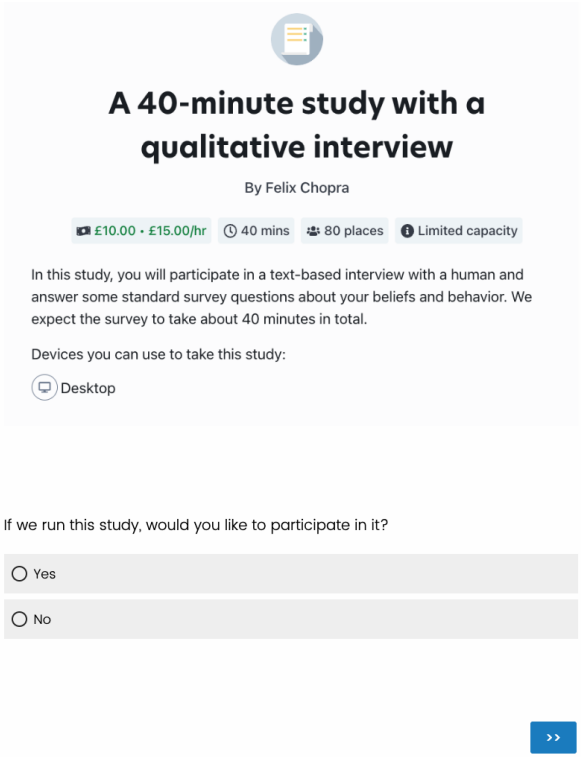
Thank you for completing the background questions in this survey. Based on your responses, you might be eligible to participate in future potential studies.

Would you be interested in participating in future studies? On the next few pages, you will find a brief description of each study. For each study described, please indicate if you are interested in receiving an invitation to participate in that study.

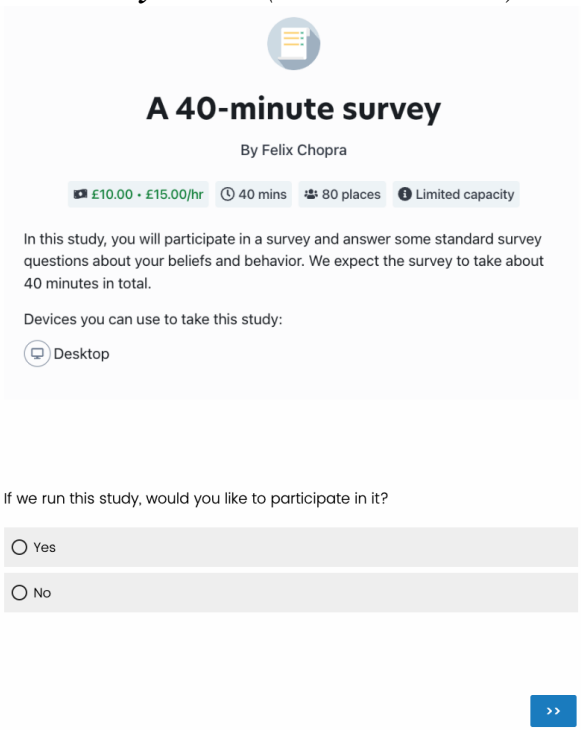
AI interview *(order randomized)*



Human interview *(order randomized)*



Survey version *(order randomized)*



F.1.6 Validation survey (May 2024)

We elicit similar sociodemographic variables as in the Screener Survey (August 2023). We do not replicate these instructions here.

To what extent do you agree or disagree with the following statement? "If you want to make any money in the stock market, you have to constantly monitor market conditions, keep up-to-date with all the news, and do a lot of research on the stocks that you plan to invest in." [Strongly disagree, Disagree, Neither agree nor disagree, Agree, Strongly agree]

To succeed with stock market investments, how often do you think it is necessary to monitor market conditions and check one's portfolio? [Less than once a year, Once a year, Once a month, Once a week, Multiple times a week, Once a day, Multiple times a day]

To succeed in the stock market, how much time do you think you would typically need to spend every week to manage your investments? [Less than 1 hour per week, 1–3 hours per week, 4–6 hours per week, 7–10 hours per week, 11–15 hours per week, More than 15 hours per week]

Which of the two statements below best describes your views?

a) The key to stock market success is to react quickly to daily market changes.
b) The key to stock market success is to ignore daily market changes.

Which of the two statements below best describes your views?

a) The key to stock market success is to actively pick a few stocks that will increase in value and avoid stocks that will decrease in value.
b) The key to stock market success is to invest broadly in all stocks without making an active selection.

Which of the following types of information do you think one would have to regularly consult when participating in the stock market to make a return on one's portfolio? Select all that apply.

a) Earnings reports (e.g., quarterly or annual financial statements of companies)
b) Market reports (e.g., stock market performance summaries and trends)
c) Forecasts from stock market analysts (e.g., buy/sell recommendations and target prices for individual stocks)
d) Company press releases (e.g., new product launches, management changes)
e) Technical analysis (e.g., charts and patterns predicting stock price movements)
f) Insider trading reports (e.g., transactions by company executives)
g) Social media and forums (e.g., investment communities, Twitter, Reddit)
h) None of the above

F.2 Tariff studies

F.2.1 Screener survey (May 2025)

Before starting, we have another quick check to ensure you're a real person who carefully reads survey instructions. Bots and automated or inattentive responses compromise our research, so thank you for your help! To confirm you're human and paying close attention, please ignore your actual state of mind for the next question.

Instead, follow the instructions below and select three response options, namely: Select the option that includes the word "relaxed." Select the third option in the list. Select the first option listed.

Which of the following describes your current state of mind?
[I feel focused, I feel tired, I feel happy, I feel low, I feel relaxed, I feel stressed, None of the above, I prefer not to answer]

(New survey page)

What is your opinion about the yearly switch to daylight saving time? Do you like or dislike it? Please use about 15 to 30 words. [text entry field]

(New survey page)

Do you approve or disapprove of the Trump administration substantially increasing tariffs on goods imported from most countries that trade with the US?
[Strongly approve, Somewhat approve, Somewhat disapprove, Strongly disapprove]

(New survey page)

Please think about the increased trade of goods and services between the US and other countries in recent decades. How do you think increased trade has affected the US?
[The US gained more than it has lost from increased trade, The US gained about as much as it has lost from increased trade, The US lost more than it has gained from increased trade]

(New survey page)

On economic policy matters, where do you see yourself on the liberal/conservative spectrum?
[Very liberal, Liberal, Moderate, Conservative, Very conservative]

(New survey page)

In politics, as of today, do you consider yourself a Republican, a Democrat, or an Independent?
[Republican, Democrat, Independent]

Who did you vote for in the 2024 presidential election between Donald Trump and Kamala Harris?
[Donald Trump, Kamala Harris, Someone else, Did not vote]

(New survey page)

(Displayed if respondent identifies as Republican)
Do you consider yourself a strong Republican or a weak Republican?
[Strong Republican, Weak Republican]

(Displayed if respondent identifies as Democrat)
Do you consider yourself a strong Democrat or a weak Democrat?
[Strong Democrat, Weak Democrat]

(Displayed if respondent identifies as Independent)
Do you lean towards the Republican Party or lean towards the Democratic Party?
[The Republican Party, The Democratic Party]

F.2.2 Tariff interview survey (June 2025)

(We elicit basic sociodemographic variables. We do not replicate these instructions here.)

In this study, you'll participate in an interview with a chatbot. This interview is a very important part of the study. Please answer thoughtfully and take your time. You have two options to answer the questions: You type the answers using your keyboard. You use the voice recording feature, see how it works in the video below. *[Short demonstration video]*

(New survey page)

Interview topic
On April 5, 2025, the United States began collecting a 10% tariff on all imports, with higher rates on some goods. On the next page, you will participate in an interview with an AI chatbot about your views about tariffs.

To ensure that the interview is as meaningful as possible, it's important that your responses provide sufficient detail. Please aim to write full sentences in your answers to allow for a meaningful conversation. There are no right or wrong answers.

We expect the interview to take about 20 minutes to complete on average.

(New survey page)

(Interview interface)

(New survey page)

How would you rate your overall experience with the interview conducted by the AI chatbot?
[Extremely positive, Very positive, Somewhat positive, Neutral, Somewhat negative, Very negative, Extremely negative]

How natural did the conversation with the AI chatbot feel?
[Extremely natural, Very natural, Somewhat natural, Neutral, Somewhat unnatural, Very unnatural, Extremely unnatural]

If you were to participate in a future study involving a qualitative interview conducted through a chat interface, would you prefer texting with the same AI chatbot or an actual human interviewer?

[Strongly prefer AI chatbot, Moderately prefer AI chatbot, Slightly prefer AI chatbot, Slightly prefer human interviewer, Moderately prefer human interviewer, Strongly prefer human interviewer]

Would you be interested in participating in an interview with an AI chatbot again? [Yes, No]

(New survey page)

How accurately did the previous interview capture how you personally think about the topic of tariffs?
[Very accurate, Somewhat accurate, Neither accurate nor inaccurate, Somewhat inaccurate, Very inaccurate]

Did the interview questions seem neutrally phrased or did the questions already suggest a certain answer?
[Very neutral, Somewhat neutral, Neither neutral nor suggestive, Somewhat suggestive, Very suggestive]

Were the interview questions open-ended, allowing you to answer in your own words, or closed-ended, limiting your response options?
[Very open-ended, Somewhat open-ended, Neither open-ended nor closed-ended, Somewhat closed-ended, Very closed-ended]

How comfortable did you feel interacting with the AI chatbot?
[Extremely comfortable, Very comfortable, Moderately comfortable, Slightly comfortable, Not at all comfortable]

(New survey page)

To what extent do you believe higher US tariffs will have positive or negative impacts on the overall US economy in the short and long term?

1. Short-term overall economic impact
2. Long-term overall economic impact

[Response options for each element above: Very negative impact, Somewhat negative impact, Neither positive nor negative impact, Somewhat positive impact, Very positive impact]

(New survey page)

Over the next 12 months, how do you expect higher US tariffs to affect the following outcomes?

1. US inflation rate
2. US real GDP growth
3. US unemployment rate
4. US trade deficit
5. Net job creation in the US
6. US stock market returns (S&P 500)
7. Variety of products available to US consumers
8. Global GDP growth

[Response options for each element above: Strongly increase, Somewhat increase, Neither increase nor decrease, Somewhat decrease, Strongly decrease]

(New survey page)

To what extent do you expect each group to gain or lose from higher US tariffs?

1. US manufacturers
2. US consumers
3. US exporters
4. High-income households
5. Low-income households

[Response options for each element above: Gain a lot, Gain a little, No effect, Lose a little, Lose a lot]

(New survey page)

How likely is it that major US trading partners will retaliate with their own tariffs?
[Very likely, Somewhat likely, Neither likely nor unlikely, Somewhat unlikely, Very unlikely]

Which statement best describes how firms will respond to higher tariffs on imported goods?
[Mostly absorb the higher cost themselves, Split the cost about 50/50 between firm and consumer, Mostly pass the cost on to US consumers]

(New survey page)

Please tell us whether you agree or disagree with the following statement: "In international trade, if one country makes more money, then it is generally the case that the other country makes less money." [Strongly agree, Agree, Neither agree nor disagree, Disagree, Strongly disagree]

(New survey page)

In which of the following industries do you work in your main job? If you are currently unemployed, think about your last job.

1. Construction
2. Durable manufacturing (lumber and wood products, furniture, metal industries, machinery and computing equipment, electrical machinery, transportation equipment, professional equipment, other durable goods)
3. Finance, insurance and real estate
4. Health and education
5. Nondurable manufacturing (food, textiles, apparel, paper products, printing and publishing, chemicals, petroleum and coal products, plastics products, leather products, other nondurable goods), agriculture and fishing, or mining
6. Retail and wholesale
7. Services (business and repair services, personal services, entertainment and recreation services, professional services and legal services)
8. Transportation, communications and other public utilities
9. Other: _____

How closely do you follow news about politics and the economy?
[Very closely, Somewhat closely, Not very closely, Not closely at all]

F.2.3 Tariff benchmark survey (July 2025)

(The survey is identical to the screener survey described in Section F.2.1 and the tariff interview study described in Section F.2.2, with the exception that we remove instructions specific to the interview. We replace the interview block with the instructions below.)

On April 5, 2025, the United States began collecting a 10% tariff on all imports, with higher rates on some goods. We will now ask you a few questions about your views on tariffs. Please aim to write full sentences. There are no right or wrong answers.

(New survey page)

What are the main reasons you support or oppose the new 10 percent tariff on imports? [text entry field]

(New survey page)

Can you think of any potential benefits or costs of tariffs that you haven't mentioned so far? [text entry field]

(New survey page)

Imagine you had to explain to someone unfamiliar with tariffs how they work and how they might affect different groups. What would you say? [text entry field]

(New survey page)

Suppose the United States suddenly raised tariffs to 30 percent on imports from all major trading partners. What do you think would happen in the short run, and what might happen further down the road? [text entry field]

(New survey page)

Is there anything else you think is important that we haven't touched on yet? [text entry field]

Is there anything you would want to know or see that could change your mind about the tariffs? [text entry field]